

Ugence Enterprise AI Governance Capability Pipeline

Repository-Based Capability Map, Development Status and Competitive Cross-Check

Define → Propose → Verify → Decide → Authorize → Clear → Execute → Assure → Measure

Agents and models may propose. Independent governance components verify, decide, authorize and clear. The runtime executes only within that authority, while assurance observes the trajectory and outcome.

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Repository-based capability map grounded in the inspected codebase snapshot. Contract-only and research-only packages are described according to their intended position while their present limitations are stated explicitly. Nothing in this document represents a production deployment, a customer validation or legal advice.

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Document control

Version	Date	Author	Change
0.1	2026-09-04	Ugence Labs	Capability map (Sections 1–15) and Appendix A use case, as supplied.
0.2	2026-09-04	Ugence Labs	Appendix B added: development status of the 45 capabilities with stage tags, evidence and canonical pipeline diagram.
0.3	2026-09-04	Ugence Labs	Body text cross-checked against package source; five claims corrected in place; competitor supplement and Appendix C added.
1.0	2026-09-04	Ugence Labs	First docx edition with cover sheet, document control, figures and copyright footer.

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1. Purpose and scope

This document explains how the enterprise AI governance capabilities found under the repository's `packages/` directory fit into one understandable operating sequence. It is intended for business leaders, governance teams, architects, risk officers and engineers.

The inspected repository snapshot contains **47 installable packages**. This document intentionally excludes the two packaged business-solution examples and covers the remaining **45 platform capabilities**. Cloud-scaling packages remain included because they are capability and integration modules that demonstrate how the general governance architecture can govern a consequential operational domain.

The package set is not uniformly production-ready. It includes implemented kernels, integration components, experimental or research-only capabilities, and contract-only foundations. A package's presence in the sequence identifies its architectural responsibility; it does not by itself establish production deployment readiness.

2. Platform operating principle

The architecture separates nine questions that should not be answered by one agent or one model:

1. **Define:** What policies, identities, boundaries and objectives govern the system?
2. **Propose:** What does the AI recommend doing, and how was the proposal prepared?
3. **Verify:** Are the proposal's claims, evidence, policy references and benchmarks trustworthy?
4. **Decide:** Has an accountable authority made the binding business and risk decision?
5. **Authorize:** Does that decision permit this exact action, actor, scope and time window?
6. **Clear:** Is the already-authorized action still safe and operationally valid immediately before execution?
7. **Execute:** How is the permitted action carried out reliably without expanding its authority?
8. **Assure:** Did execution remain within its authority, and did the intended real-world effect occur?
9. **Measure:** Was the governed system ready, and what value did the authorized action create?

The central rule is:

Agents and models may propose. Independent governance components verify, decide, authorize and clear. The runtime executes only within that authority, while assurance observes the trajectory and outcome.

3. Layman architecture sequence

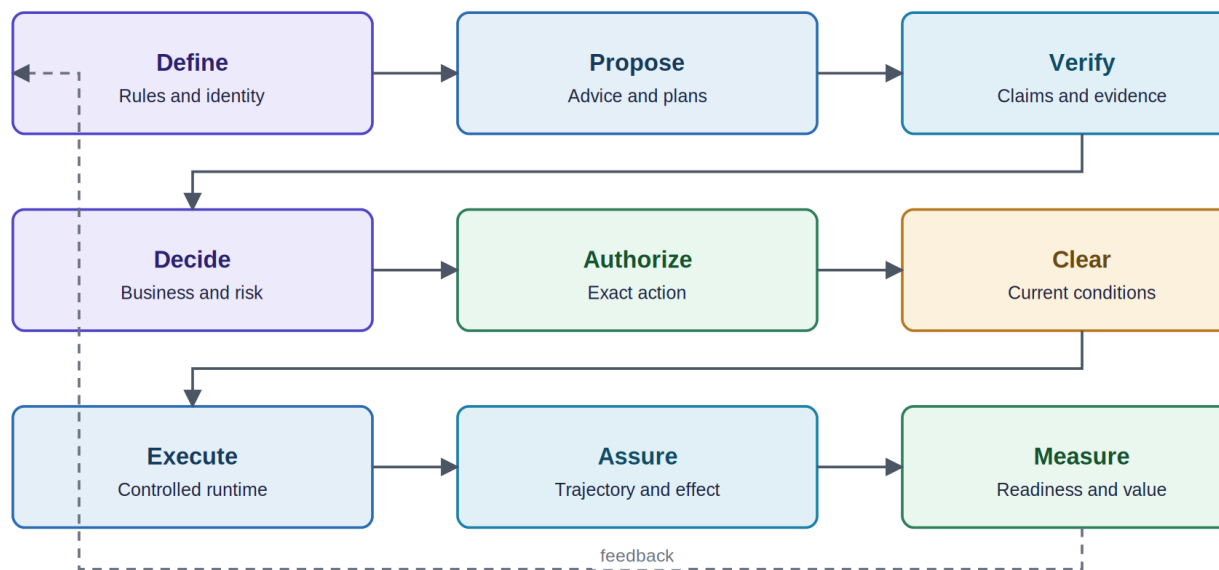


Figure 1. The nine-stage governance sequence with its feedback loop.

Three shared foundations support the whole sequence:

- **Common contracts and identity:** all components use stable, machine-verifiable meanings.
- **Provider and integration boundaries:** modules can be composed without collapsing their responsibilities.
- **Fail-closed authority separation:** absence, ambiguity, expiry or mismatch does not silently become permission.

4. Shared foundations across the sequence

These packages do not represent a single business step. They provide the common language, identities and interoperability needed across multiple steps.

1. Governance Contracts

Package: `packages/governance-contracts`

Sequence alignment: Cross-cutting foundation for **Verify** → **Decide** → **Authorize** → **Execute** → **Assure**.

Pipeline role: Defines provider-neutral request, result, evidence, execution and system-binding contracts so independently built capabilities exchange the same typed facts without inheriting one another's authority.

Why it is necessary:

1. Prevents each governance module from inventing incompatible meanings for actions, assertions, evidence and execution results.
2. Preserves authority boundaries by making the contracts neutral rather than embedding policy or decision power in shared data structures.
3. Enables audit, replay and substitution of providers because requests and results have stable, testable shapes.

2. Governance Provider Framework

Package: `packages/governance-provider-framework`

Sequence alignment: Cross-cutting composition layer for **Verify** → **Authorize** → **Execute**.

Pipeline role: Supplies provider registry, resolution, lifecycle and conformance mechanics through which

evidence, action-governance and execution providers can be discovered and invoked without the framework becoming an authority.

Why it is necessary:

1. Allows TAP, ActionGate and future providers to plug into a common mechanism while remaining independent peers.
2. Separates provider operations from the governance meaning owned by each capability.
3. Makes provider compatibility, health and failure handling explicit instead of burying them inside application code.

3. JSON Canonicalization Scheme

Package: `packages/jcs`

Sequence alignment: Identity foundation across **Define** → **Propose** → **Verify** → **Authorize** → **Assure**.

Pipeline role: Produces deterministic canonical bytes and hashes for already-parsed data, allowing the platform to prove that different components are referring to the exact same policy, proposal, evidence object or action representation.

Why it is necessary:

1. Makes tampering or accidental semantic drift mechanically detectable through stable digests.
2. Supports exact-action authorization because logically identical inputs receive reproducible identities.
3. Enables cross-system verification and replay without depending on incidental JSON formatting or field order.

4. Benchmark Registry Contracts

Package: `packages/benchmark-registry`

Sequence alignment: Foundation for **Define** → **Verify** → **Measure**.

Pipeline role: Defines exact, digest-bound benchmark identities, lifecycle vocabulary and typed refusals; it makes floating references such as an unspecified “latest benchmark” structurally avoidable. The current package is contract-only and does not itself operate a registry.

Why it is necessary:

1. Prevents favorable benchmarks from being silently substituted across tenants, geographies, populations, metrics or time periods.
2. Gives readiness and value assessments an exact comparison target that can later be independently resolved and audited.
3. Separates benchmark identity from benchmark trust, avoiding the false assumption that possessing a benchmark makes it valid.

5. UVI Policy Contracts

Package: `packages/uvi-policy-contracts`

Sequence alignment: Foundation for **Define** → **Measure**.

Pipeline role: Defines immutable policy and assessment-context shapes for Ugence Value Intelligence, including policy references, thresholds, gates, intended outcomes, geography, domain, valuation and readiness context. It creates no policy authority by itself.

Why it is necessary:

1. Ensures value and readiness evaluations are tied to an explicit business, geographic and intended-outcome context.
2. Prevents caller-controlled value multipliers and floating policy references from gaming an assessment.
3. Gives Policy Authority and downstream evaluators a stable contract boundary without merging policy issuance with measurement.

5. Define — establish policies, identities and boundaries

6. Policy Authority

Package: `packages/policy-authority`

Sequence alignment: **Define**, with continuing control over **Verify** and **Authorize** through trusted resolution and revocation.

Pipeline role: Issues, signs, registers, resolves and revokes versioned policies used across the platform. It establishes which exact policy artifact is currently trusted without deciding a business case or executing an action.

Why it is necessary:

1. Converts policy from editable prose or configuration into signed, versioned and revocable governance artifacts.
2. Prevents agents and applications from selecting whichever policy version is most convenient at runtime.
3. Provides a common platform authority for multiple policy families while keeping business decisions and execution authority separate.

7. Policy Workflow Compiler

Package: `packages/tooling/policy-workflow-compiler`

Sequence alignment: **Define** → **Propose**.

Pipeline role: Compiles a reviewed structured policy pack into deterministic Workflow IR, assurance specifications, capability requirements, audit schemas and content-addressed artifacts. It describes how governance must be composed but does not exercise authority.

Why it is necessary:

1. Translates human-approved policy into a machine-readable workflow without relying on runtime prompt interpretation.
2. Makes governance structure reproducible, diffable and testable before an agent operates.
3. Creates the workflow and assurance artifacts consumed by proposers, workforce planning and downstream governance controls.

8. Agent Constitution Policy

Package: `packages/integration/agent-constitution-policy`

Sequence alignment: **Define**.

Pipeline role: Introduces an issuable Policy Authority family that states the signed structural bounds of governed roles, including permitted candidate dispositions, review actions and tool scopes.

Why it is necessary:

1. Gives each governed role a durable constitutional ceiling that cannot be expanded by the agent itself.
2. Separates organizational role constraints from transient prompts and model behavior.
3. Makes role bounds signable, versioned, resolvable and revocable through the shared Policy Authority.

9. Agent Constitution Activation

Package: `packages/integration/agent-constitution-activation`

Sequence alignment: **Define** → **Propose**.

Pipeline role: Provides the composition root, preflight checks, governed reference-map derivation and key-material-free receipts needed to issue and activate a constitution through Policy Authority.

Why it is necessary:

1. Turns a constitution definition into an explicitly activated, deployment-consumable governance relationship.
2. Ensures configuration and reference mappings are checked before governed proposals are produced.
3. Keeps key material out of activation receipts while retaining evidence that the activation path was completed.

10. Cloud Scaling Capacity-Bounds Policy

Package: `packages/integration/cloud-scaling-capacity-bounds-policy`

Sequence alignment: Define → Authorize.

Pipeline role: Defines the Policy Authority family and adapter for signed, digest-bound cloud capacity ceilings that later constrain scaling recommendations and authorization candidates.

Why it is necessary:

1. Encodes business or operational capacity limits as governed policy rather than controller tuning.
2. Prevents a scaling recommender or executor from unilaterally increasing its permitted range.
3. Demonstrates how domain-specific constraints attach to a shared enterprise policy authority.

11. Agentic Proposer Strategy-Permission Policy

Package: `packages/integration/agentic-proposer-strategy-permission-policy`

Sequence alignment: Define → Propose.

Pipeline role: Defines an issuable Policy Authority family specifying which declared reasoning strategies a governed role is permitted to use. Permission constrains proposal preparation but grants no compute, tool or execution authority.

Why it is necessary:

1. Lets an organization govern reasoning procedures independently of a model's self-selected behavior.
2. Makes strategy permission signed, versioned and revocable rather than a mutable local setting.
3. Prevents permission to reason in a certain way from being confused with permission to act.

6. Propose — prepare advice, plans and candidates

12. Agent Constitution Conformance

Package: `packages/integration/agent-constitution-conformance`

Sequence alignment: Define → Propose → Verify.

Pipeline role: Resolves an issued constitution through Policy Authority and checks whether presented role facts remain within its signed structural bounds. It returns structural conformance, not an operational permission or denial.

Why it is necessary:

1. Makes the constitution enforceably checkable rather than merely issuable documentation.
2. Fails closed when the exact constitution, tenant, role binding, approval or lifecycle state cannot be trusted.
3. Prevents a proposal from silently relying on role declarations broader than the organization authorized.

13. Agentic Proposer

Package: `packages/capabilities/agentic-proposer`

Sequence alignment: Propose.

Pipeline role: Produces identity-bound advisory candidates and a structured advisory using bounded context, role contracts, observations, eligibility and declared reasoning strategy. It may propose, request evidence, abstain or escalate; it does not decide or authorize.

Why it is necessary:

1. Converts opaque agent intent into a typed, inspectable proposal that governance components can evaluate.
2. Preserves the crucial separation between generating options and granting permission to act.
3. Creates deterministic identities and replay checks around proposal artifacts even though private model reasoning is not exposed or claimed deterministic.

14. Strategy-Permission Runtime Resolver

Package: `packages/integration/agent-proposer-strategy-permission-runtime`

Sequence alignment: Define → Propose → Verify.

Pipeline role: Resolves the exact signed strategy-permission policy for the proposer through Policy Authority and returns permitted strategy tokens only after tenant, scope, reference, approval and lifecycle checks succeed.

Why it is necessary:

1. Bridges the deliberately separate Proposer and Policy Authority packages without causing either to absorb the other's responsibility.
2. Prevents an agent from treating an unsigned configuration value as governed reasoning permission.
3. Ensures expired, revoked, mismatched or unresolved strategy policy produces no degraded permissive answer.

15. Agent Workforce Composer

Package: `packages/capabilities/agent-workforce-composer`

Sequence alignment: Define → Propose.

Pipeline role: Converts Workflow IR roles into capability requirements, determines eligible agents under hard constraints, ranks eligible candidates, composes a bounded team and proposes least-privilege permissions and fallbacks. It grants and schedules nothing.

Why it is necessary:

1. Ensures agent assignment is based on explicit capability and enterprise constraints rather than informal selection.
2. Produces complete explanations for both selected and eliminated candidates.
3. Limits proposed team permissions before runtime governance, reducing the authority surface that later stages must evaluate.

16. Reasoning Method Governance

Package: `packages/capabilities/reasoning-method-governance`

Sequence alignment: Define → Propose → Measure.

Pipeline role: Supplies research-only shared contracts for reasoning-method catalogs, task classes, execution records, fit assessments, evidence views, comparison requests and research plans. It defines the vocabulary but performs no comparison or approval.

Why it is necessary:

1. Creates a stable language for comparing reasoning methods without coupling contracts to an experimental runtime.
2. Separates observed telemetry, evidence status and fit assessment so none is mistaken for authority.
3. Allows advisor, comparison and pilot packages to evolve independently while remaining interoperable.

17. Reasoning Method Advisor

Package: `packages/capabilities/reasoning-method-advisor`

Sequence alignment: Propose.

Pipeline role: Provides research-only, deterministic, rule-derived advice about which reasoning methods

qualify for a governed task profile. It explains inclusions, exclusions and trade-offs and names a primary method only when exactly one qualifies.

Why it is necessary:

1. Makes reasoning-method selection explainable and replayable rather than an ungoverned model preference.
2. Preserves ambiguity by returning zero, one or many qualifying methods instead of manufacturing a winner.
3. Establishes a design-time control point for matching reasoning approach to consequence, reversibility and task characteristics.

18. Readiness Comparison

Package: `packages/capabilities/readiness-comparison`

Sequence alignment: **Propose** → **Measure**.

Pipeline role: Executes a research-only pure comparison over reasoning-method fit and resource outcomes. It does not fetch benchmarks, infer authority or create approval-bearing results.

Why it is necessary:

1. Separates empirical comparison from the advisor's rule-derived recommendation.
2. Makes comparison logic deterministic, timestamp-explicit and reproducible.
3. Provides evidence that can improve future method choice without prematurely converting research results into deployment authority.

19. Trusted Workflow-Fit Pilot

Package: `packages/capabilities/workflow-fit-pilot`

Sequence alignment: **Propose** → **Verify** → **Measure**.

Pipeline role: Runs a research-only preregistered workflow-fit study with separate-process capture, recomputed telemetry, declared evaluator scoring, comparison and lineage tracking. Its current judgments remain explicitly unverified and non-approval-bearing.

Why it is necessary:

1. Tests whether recommended reasoning methods actually fit representative workflows under a declared study design.
2. Reduces self-reporting risk by capturing and recomputing telemetry at a controlled boundary.
3. Creates an evidence path from design-time advice toward future governed readiness without overstating research maturity.

20. Model Selection

Package: `packages/capabilities/model-selection`

Sequence alignment: **Define** → **Propose**.

Pipeline role: Selects an eligible model and provider from an approved set under deterministic policy constraints. It owns eligibility and selection, not request routing or provider execution. The package now brands itself **Model Authority**: its canonical public contract is a `ModelAuthorizationDecision` with dispositions `ALLOW`, `DENY`, `HOLD` and `ESCALATE`, and the selection-era names survive only as deprecated aliases.

Why it is necessary:

1. Prevents agents from choosing unapproved models solely for convenience, performance or cost.
2. Makes model eligibility a governed, replayable decision rather than a hidden runtime heuristic.
3. Separates approved model choice from the operational mechanics of routing and invoking it.

21. LLM Steering Controller

Package: `packages/capabilities/llm-steering-controller`

Sequence alignment: **Propose.**

Pipeline role: Produces advisory LLM-routing recommendations through candidate discovery, hard-constraint filtering, scoring and fallback or escalation guidance. It does not invoke providers.

Why it is necessary:

1. Allows performance, cost, latency and policy constraints to be considered before an LLM call is routed.
2. Keeps routing recommendations explainable and independent of provider execution.
3. Provides controlled fallbacks and escalation when no candidate safely satisfies the request.

22. Context Minimization

Package: `packages/capabilities/context-minimization`

Sequence alignment: **Propose** → **Execute** → **Assure.**

Pipeline role: Deterministically removes unnecessary context while protecting required units and preserving a caller-defined equivalence condition, with token accounting and fail-closed behavior.

Why it is necessary:

1. Reduces sensitive-data exposure by limiting what an agent or model receives.
2. Lowers token cost and latency without treating compression as acceptable when required meaning would change.
3. Produces measurable context and token records for later budget assurance and governance audit.

23. StoryGraph

Package: `packages/capabilities/storygraph`

Sequence alignment: **Propose** → **Verify.**

Pipeline role: Analyzes proposed event or action sequences for deterministic, policy-defined sequence risk and emits advisory evidence.

Why it is necessary:

1. Detects harms that emerge from a sequence of individually acceptable steps.
2. Supplies structured risk evidence before a binding decision or authorization is made.
3. Makes temporal and causal workflow patterns inspectable instead of evaluating every action in isolation.

24. Cloud Scaling Controller

Package: `packages/capabilities/cloud-scaling-controller`

Sequence alignment: **Propose.**

Pipeline role: Converts normalized workload observations into deterministic, explainable and provider-neutral scaling recommendations. It contains no actuation capability.

Why it is necessary:

1. Demonstrates a real operational proposer that remains strictly separate from authorization and execution.
2. Produces explainable capacity recommendations and evidence for risk evaluation.
3. Allows shadow evaluation and replay before the organization permits live infrastructure changes.

7. Verify — establish trust in claims, evidence and references

25. TAP Assertion Governance Provider

Package: `packages/providers/tap`

Sequence alignment: **Verify.**

Pipeline role: Evaluates whether a material assertion is supported, unsupported, constrained or indeterminate by supplied evidence; the provider's native outcome set carries a fifth **UNKNOWN** state that is mapped fail-safe to indeterminate. TAP governs claims in the assessment path and never authorizes an action.

Why it is necessary:

1. Stops recommendations from being treated as trustworthy merely because an AI stated them confidently.
2. Preserves uncertainty and infrastructure failure as indeterminate rather than promoting them to support.
3. Keeps evidence judgment independent from ActionGate's action-authorization responsibility.

26. Trusted Evidence Authority

Package: `packages/trusted-evidence-authority`

Sequence alignment: **Verify** → **Assure**.

Pipeline role: Defines canonical evidence identity and verifies trust anchors, signatures, key entitlement, revocation and scope before issuing signed evidence-verification receipts. A receipt proves verification under configured trust but authorizes nothing.

Why it is necessary:

1. Distinguishes evidence authenticity and provenance from the truth of a claim or permission to act.
2. Prevents signatures from being trusted without checking key entitlement, validity window and revocation.
3. Creates independently re-verifiable receipts that downstream risk and audit components can rely upon.

27. Benchmark Registry Authority

Package: `packages/benchmark-registry-authority`

Sequence alignment: **Verify** → **Measure**.

Pipeline role: Defines registry lifecycle, exact resolution, trust-anchor and verified-result contracts above benchmark identity. In the inspected snapshot it remains a contract and non-authoritative lifecycle surface: it has no operational store, verifier, clock or authority-issued result.

Why it is necessary:

1. Establishes how a benchmark can eventually be admitted, resolved, revoked and trusted without weakening exact identity.
2. Prevents benchmark possession or retrieval from being confused with verified validity.
3. Provides the authority boundary required for credible readiness, comparison and value measurement.

28. Risk Authority Evidence Runtime

Package: `packages/integration/risk-authority-evidence-runtime`

Sequence alignment: **Verify** → **Decide**.

Pipeline role: Composes trusted evidence admission and TAP-based control assurance to produce trusted control results for Risk Authority.

Why it is necessary:

1. Converts raw evidence references into control results that have passed explicit trust and assertion checks.
2. Prevents Risk Authority from relying directly on caller-asserted evidence quality.
3. Preserves non-compensatory control semantics so one passing control cannot erase a mandatory failure.

29. Cloud Scaling Producer Attestation

Package: `packages/integration/cloud-scaling-producer-attestation`

Sequence alignment: **Verify**.

Pipeline role: Verifies the authenticity of the producer that created a cloud-scaling recommendation and binds that attestation to an authorization candidate. The attestation grants no authority.

Why it is necessary:

1. Prevents an untrusted or substituted controller from injecting a recommendation into the governance path.
2. Binds provenance to the exact candidate rather than trusting a generic producer label.
3. Separates producer authenticity from policy validity, risk acceptance and action authorization.

30. Cloud Scaling Policy Authenticity

Package: `packages/integration/cloud-scaling-policy-authenticity`

Sequence alignment: **Define** → **Verify** → **Authorize**.

Pipeline role: Verifies that the capacity policy attached to a scaling candidate is the exact trusted Policy Authority artifact expected for the tenant, scope and time. Its proof grants nothing by itself.

Why it is necessary:

1. Prevents an expired, revoked, mismatched or fabricated capacity ceiling from governing a live action.
2. Keeps policy authenticity distinct from the decision that the proposed scaling action is acceptable.
3. Makes the domain policy reference independently auditable before exact-action authorization.

8. Decide — make accountable business and risk determinations

31. Decision Authority

Package: `packages/capabilities/decision-authority`

Sequence alignment: **Decide**.

Pipeline role: Owns the bounded binding business decision. It determines whether an accountable, delegated authority approves the governed case; it does not execute or replace exact-action authorization.

Why it is necessary:

1. Ensures consequential business outcomes are owned by an explicit accountable authority rather than an advisory agent.
2. Records the scope, conditions and validity of the binding decision for downstream enforcement.
3. Prevents a model recommendation, evidence result or risk score from silently becoming organizational approval.

32. Risk Authority

Package: `packages/risk_authority`

Sequence alignment: **Verify** → **Decide** → **Authorize**.

Pipeline role: Applies non-compensatory control evaluation and converts an approved governance decision into cryptographically bound, scoped, time-bound and revocable machine authority for exact-action enforcement. Envelope issuance, Ed25519 signing, expiry and revocation are implemented on the reference path; with `production_mode` set, issuance and exact-action authorization fail closed as unimplemented Phase 5 work, so production integration currently stops at a non-executable `RiskDecision`.

Why it is necessary:

1. Bridges human or organizational approval to machine-enforceable authority without widening the original decision.
2. Fails closed when mandatory controls are missing, stale, unknown or failed.
3. Makes runtime authority scoped, expiring and revocable rather than a permanent blanket permission.

33. Cloud Scaling Risk Integration

Package: `packages/integration/cloud-scaling-risk-integration`

Sequence alignment: **Propose** → **Decide**.

Pipeline role: Projects an advisory cloud-scaling recommendation into a Risk Authority subject-risk evaluation without executing or authorizing the recommendation.

Why it is necessary:

1. Translates domain-specific recommendation facts into the common enterprise risk model.
2. Preserves one-way dependency so the scaling controller remains advisory and unaware of authority internals.
3. Allows capacity, workload and operational risks to influence governance before infrastructure mutation.

9. Authorize — permit one exact consequential action

34. Cloud Scaling Authorization Contracts

Package: `packages/integration/cloud-scaling-authorization-contracts`

Sequence alignment: **Propose** → **Verify** → **Authorize**.

Pipeline role: Binds a scaling recommendation, risk result, producer identity and relevant references into a non-authoritative capacity-action candidate that downstream authority can evaluate. The candidate itself grants nothing.

Why it is necessary:

1. Creates one exact object linking the recommendation to the evidence and risk context being authorized.
2. Prevents downstream authorization from acting on an ambiguous or reconstructed version of the proposal.
3. Maintains the distinction between an authorization candidate and actual machine authority.

35. Risk Authority Runtime Composition

Package: `packages/integration/risk-authority-runtime`

Sequence alignment: **Decide** → **Authorize**.

Pipeline role: Provides a fail-closed composition of Risk Authority, the canonical Decision Authority and ActionGate so the binding decision, machine-authority envelope and exact action are evaluated together.

Why it is necessary:

1. Connects separately owned authorities without merging their responsibilities into one opaque engine.
2. Ensures an action cannot bypass the chain from business decision to scoped authority to exact-action check.
3. Centralizes fail-closed integration behavior at the point where mismatches would otherwise become dangerous.

36. ActionGate

Package: `packages/providers/actiongate`

Sequence alignment: **Authorize**.

Pipeline role: Evaluates whether the exact proposed action is authorized by the supplied authority, policy, risk, evidence and decision context. It returns one of four outcomes through the neutral action-

governance contract (authorized, authorized with constraints, denied, indeterminate) but never dispatches or executes the action.

Why it is necessary:

1. Stops a valid general decision from being reused for a different action, resource, scope or actor.
2. Provides a deterministic enforcement point immediately before operational clearance and execution.
3. Preserves separation between authorization and actuation, limiting the consequences of an ActionGate defect or integration error.

10. Clear — recheck present conditions immediately before execution

37. Action Clearance

Package: `packages/capabilities/action-clearance`

Sequence alignment: Clear.

Pipeline role: Evaluates whether an already-authorized exact action remains operationally clear under trusted current-state signals. It may preserve, narrow, hold, escalate or block existing authority, but can never create or broaden it.

Why it is necessary:

1. Handles the gap between authorization time and execution time, when environment conditions may change.
2. Stops stale authorization from overriding freezes, incidents, conflicts, expired dependencies or other current constraints.
3. Adds a last safe checkpoint without duplicating Decision Authority or ActionGate.

11. Execute — perform only the permitted operation

38. Agent Runtime

Package: `packages/runtime/agent-runtime`

Sequence alignment: Propose → Authorize → Execute → Assure.

Pipeline role: Coordinates task and workflow lifecycles, provider invocation, retry, timeout, cancellation, budgets, concurrency, checkpoints and recovery. Consequential transitions cross an external governance boundary and fail closed when governance is not configured.

Why it is necessary:

1. Provides a canonical execution state and controlled lifecycle rather than letting every agent improvise orchestration.
2. Keeps governance checks inside the indivisible transition from proposal to exact provider action.
3. Produces durable execution and telemetry records needed for recovery, audit and post-action assurance.

39. Cloud Scaling Operations

Package: `packages/capabilities/cloud-scaling-operations`

Sequence alignment: Execute.

Pipeline role: Performs controlled Kubernetes or ArgoCD scaling operations after external authorization, with dry-run as the default of four execution modes (dry-run, simulation, shadow, live). It is the domain actuation layer and does not create its own authority.

Why it is necessary:

1. Demonstrates how governance reaches a real consequential infrastructure mutation.
2. Prevents the advisory controller from containing hidden execution capability.

3. Supports safer adoption through dry-run behavior and explicit authorization-gated mutation.

12. Assure — monitor authority, trajectory and real-world effect

40. Risk Authority Status Runtime

Package: `packages/integration/risk-authority-status-runtime`

Sequence alignment: Authorize → Assure.

Pipeline role: Manages post-issuance machine-authority lifecycle, including revocation and epoch propagation, around the Risk Authority authorization artifact.

Why it is necessary:

1. Ensures issued authority can be invalidated after approval when policy, risk or organizational conditions change.
2. Propagates revocation state so distributed consumers do not continue using stale authority.
3. Treats authorization as a living lifecycle object rather than a one-time permanent token.

41. Risk Authority Runtime Assurance

Package: `packages/integration/risk-authority-runtime-assurance`

Sequence alignment: Execute → Assure → Decide.

Pipeline role: Observes runtime trajectory and can cause previously valid machine authority to be reassessed through the post-issuance intake. It observes and assesses but never mints, widens or mutates authority itself.

Why it is necessary:

1. Detects when actual execution behavior drifts from the trajectory assumed during authorization.
2. Creates a governed feedback route from runtime observations back to authority reassessment.
3. Prevents an initially valid authorization from remaining unquestioned throughout a changing workflow.

42. Risk Authority Execution Assurance

Package: `packages/integration/risk-authority-execution-assurance`

Sequence alignment: Execute → Assure → Decide.

Pipeline role: Reconciles the authorized action with the observed execution and real-world effect, composes Decision Authority reconciliation, and can trigger reassessment of previously valid authority.

Why it is necessary:

1. Distinguishes “the command ran” from “the authorized and intended effect occurred.”
2. Detects partial, divergent, failed or unexpected effects that pre-action checks cannot see.
3. Closes the governance loop by turning post-effect evidence into a reason for renewed risk and decision evaluation.

43. Context-Minimization Token-Accounting Runtime

Package: `packages/integration/context-minimization-token-accounting-runtime`

Sequence alignment: Execute → Assure → Measure.

Pipeline role: Converts Agent Runtime provider-attempt telemetry into Context Minimization accounting records and settles runtime budgets using measured token usage.

Why it is necessary:

1. Replaces caller estimates with observed consumption when enforcing shared runtime budgets.
2. Connects context minimization to measurable cost, efficiency and governance outcomes.
3. Supports tenant isolation, reconciliation and audit of token use across attempts and workflows.

13. Measure — evaluate readiness and governed value

44. Agent Value Readiness

Package: packages/capabilities/agent-value-readiness

Sequence alignment: Define → Verify → Measure.

Pipeline role: Produces a deterministic, advisory and non-financial readiness determination across intelligence fitness, capability readiness and adoption readiness, under governed policy and system-binding context.

Why it is necessary:

1. Tests whether an agent is ready for an intended business outcome before jumping directly to ROI claims.
2. Keeps readiness multidimensional so strength in one area cannot automatically compensate for a mandatory weakness elsewhere.
3. Separates advisory readiness from deployment authorization, preserving accountability for the actual deployment decision.

45. Governed Value

Package: packages/governed-value

Sequence alignment: Measure → Define.

Pipeline role: Calculates net governed value per governed action along explicit evidence, authority and outcome classification axes. The current kernel scores caller-reported inputs, fixes every result at **REPORTED** evidence status and **UNVERIFIED** authority status, and does not yet bind to an authorization artifact. It is designed to measure value after governance rather than treat an unverified forecast as realized ROI.

Why it is necessary:

1. Connects governance controls to measurable business value instead of presenting governance only as compliance overhead.
2. Distinguishes reported, forecast and observed outcomes so weak evidence is not promoted into financial truth.
3. Feeds cost, loss, benefit and authorization outcomes back into policy revision and future investment decisions.

14. End-to-end interpretation

The 45 capabilities form four complementary layers:

1. **Governance definition layer:** establishes policies, constitutions, identities, benchmarks and machine-readable workflow rules.
2. **Agent intelligence layer:** prepares candidates, teams, reasoning-method advice, model selection, context and domain recommendations without owning authority.
3. **Authority and execution layer:** verifies claims and evidence, makes the binding decision, mints bounded machine authority, checks the exact action, clears current conditions and executes.
4. **Assurance and value layer:** observes authority status, runtime trajectory, real-world effect, resource consumption, readiness and governed value.

The platform's differentiation is therefore not any single gate. It is the **non-collapsible chain of accountability**:

```

Policy authors define what is allowed
    ↓
Agents prepare inspectable proposals
    ↓
Evidence authorities establish what can be trusted
    ↓
  
```

```

Decision authority owns the binding business outcome
↓
Risk authority converts that decision into bounded machine authority
↓
ActionGate matches authority to the exact action
↓
Action Clearance checks the present operational world
↓
Agent Runtime and domain operations execute without expanding authority
↓
Assurance reconciles trajectory and real-world effect
↓
Readiness and governed value inform the next policy cycle

```

15. Why the complete pipeline is necessary

1. **No self-authorization:** the component proposing an action is not the component permitted to approve or execute it.
2. **No trust by assertion:** claims, evidence, policies, benchmarks and producer identity have distinct verification paths.
3. **No floating authority:** decisions and permissions are tied to exact identities, versions, scopes, actors, times and actions.
4. **No one-time governance:** revocation, runtime trajectory, effects, budgets and outcomes remain governed after authorization.
5. **No compliance-value trade-off:** readiness and governed-value measurement show whether controlled execution produces defensible business outcomes.

Repository interpretation note: This document describes the architectural responsibility visible in each package's metadata, README and boundaries in the inspected codebase snapshot. Contract-only and research-only packages are described according to their intended position while their present limitations are stated explicitly.

Appendix A — Enterprise Use Case: Governed Autonomous Cloud Scaling

A.1 Purpose of the appendix

This appendix tests the architecture against one concrete enterprise situation rather than assuming that all 45 capabilities must participate in every transaction. Its purpose is to show:

- which capabilities are essential to the live governance path;
- which are specifically required by the cloud-scaling domain;
- which improve safety, efficiency or governance depth but are optional for an initial deployment;
- which belong to design-time evaluation rather than the live action path; and
- which are unnecessary when the use case uses one preassigned scaling agent and a fixed execution method.

The scenario is illustrative. Names, thresholds, times and financial figures are examples, not values found in the repository or claims about a real deployment.

A.2 Scenario narrative

A.2.1 Enterprise setting

Northstar Digital Bank operates a regulated customer-facing payments platform on Kubernetes. The service normally runs 120 application replicas across two availability zones. A product launch and a national sales event are expected to produce a sharp traffic increase between 18:00 and 22:00 UTC.

The bank wants AI-assisted capacity management because manual scaling is slow, reactive and expensive. However, it does not want a forecasting model or operations agent to possess unrestricted production credentials or the ability to convert its own recommendation into an infrastructure change.

The organization establishes these illustrative requirements:

- maintain payment API latency below the governed service objective;
- preserve an operational reserve in both availability zones;
- never exceed 200 replicas without separate executive approval;
- restrict ordinary autonomous changes to a maximum increase of 60 replicas;
- require trusted workload observations and a known recommendation producer;
- refuse autonomous mutation during a declared change freeze or active severity-one incident;
- require the binding business decision and machine authorization to expire after a short window;
- execute only the exact authorized Kubernetes or ArgoCD operation; and
- verify both that the command completed and that capacity, latency and error-rate effects matched the authorized intent.

A.2.2 Define: governance is established before the traffic spike

Operations, finance, security and risk owners first agree on a structured scaling policy. Policy Authority issues the approved capacity-bounds policy as a signed, versioned and revocable artifact. The Policy Workflow Compiler translates the reviewed policy pack into deterministic Workflow IR and assurance requirements.

The policy states the permitted replica range, environment, service identity, validity interval, approval requirements, required evidence and escalation conditions. The scaling agent may recommend capacity changes, but its role does not include business approval, machine-authorization issuance or direct infrastructure mutation.

If the deployment adopts Agent Constitution controls, the role is additionally bound to a signed constitution limiting its candidate dispositions, review actions and tool scopes. Constitution activation and conformance then establish that the presented role remains inside those bounds. These controls are valuable for a reusable multi-agent estate, but they are not indispensable to a first single-agent scaling pilot when equivalent identity and role restrictions are already imposed by the deployment environment.

A.2.3 Propose: the AI prepares an inspectable recommendation

At 17:42 UTC, the Cloud Scaling Controller consumes normalized workload and infrastructure observations. CPU utilization, memory pressure, request queue depth and latency trend indicate that 120 replicas are unlikely to maintain the governed service objective during the predicted surge.

The controller recommends increasing capacity from 120 to 180 replicas. It produces an explanation and component evidence but performs no actuation. Agentic Proposer can wrap the domain recommendation into a structured advisory bound to the agent role, context, observations, candidate set, declared strategy and constitution reference where configured.

Context Minimization may reduce the operational context sent to an LLM or agent while protecting the policy, current capacity, change-freeze state and critical service signals. Model Selection and LLM Steering are useful only when the workflow dynamically chooses among models or providers. If the scaling controller is deterministic and the deployment fixes the model/provider, these two packages are optional.

Agent Workforce Composer is not required for the minimum scenario because one preassigned scaling agent owns proposal preparation. It becomes relevant when separate forecasting, cost, reliability and operations agents must be selected and composed into a governed team. The reasoning-method governance, advisor, comparison and workflow-fit pilot packages likewise belong to design-time research: they can help determine how the agent should reason, but they should not delay every live scaling transaction.

A.2.4 Verify: the platform asks whether the recommendation can be trusted

TAP evaluates the material claims behind the recommendation: whether the supplied observations support the forecasted capacity shortfall and whether the recommended increase is constrained or indeterminate. Missing evidence, malformed results or provider failure must not become “supported.”

Trusted Evidence Authority can validate evidence identity, signatures, key entitlement, validity and revocation before producing independently verifiable receipts. Cloud Scaling Producer Attestation checks that the recommendation came from the expected producer. Cloud Scaling Policy Authenticity verifies that the cited capacity policy is the exact trusted Policy Authority artifact for this tenant, scope and time.

The Benchmark Registry packages are useful if the bank compares the recommendation or later results against formally governed workload, reliability or cost benchmarks. They are not required for the immediate live action when the decision relies on current trusted observations and literal governed thresholds. The inspected Benchmark Registry Authority package is also not yet an operational authoritative registry, so it must not be represented as one.

A.2.5 Decide: an accountable authority accepts or refuses the business risk

Risk Authority Evidence Runtime turns admitted evidence and control-assurance results into trusted control inputs. Cloud Scaling Risk Integration projects the recommendation into the enterprise risk case without authorizing or executing it.

Decision Authority then owns the binding business determination: for example, approve the increase to 180 replicas for the production payments service, subject to the stated policy, evidence, time window and conditions. The model recommendation is not the decision. TAP support is not the decision. A risk score is not the decision.

Risk Authority evaluates mandatory controls non-compensatorily. A strong forecast cannot compensate for a failed change-freeze control, an untrusted producer or a missing required approval. If controls and the binding decision are valid, Risk Authority creates scoped, time-bound and revocable machine authority that cannot exceed the decision.

A.2.6 Authorize: the exact proposed mutation must match the authority

Cloud Scaling Authorization Contracts bind the exact recommendation, risk result, producer and policy references into a capacity-action candidate. Risk Authority Runtime composes the canonical Decision Authority, Risk Authority and ActionGate path.

ActionGate checks the exact action: the authorized actor, service, environment, operation, present replica count, target replica count, constraints and validity window. Authority to increase the payments service from 120 to 180 replicas cannot be reused to scale another service, change a different environment, increase to 240 replicas or perform an unrelated infrastructure operation.

An **AUTHORIZED** outcome (the provider's native **ALLOW**, mapped through the neutral action-governance contract alongside **AUTHORIZED_WITH_CONSTRAINTS**, **DENIED** and **INDETERMINATE**) means the exact action matched the supplied authority and context. It does not mean the operation has been dispatched, remains safe under changing conditions or has succeeded.

A.2.7 Clear: the platform rechecks the live operational world

Seconds before execution, Action Clearance evaluates trusted current-state signals. It checks whether a change freeze began after authorization, a severity-one incident is active, the service state changed, the authorization expired or another controller already altered capacity.

If the action remains valid, clearance is **CLEAR**. If information is temporarily incomplete, it can be held. If a human decision is needed, it can be escalated. If a prohibiting condition exists, it is blocked. Action Clearance can narrow or stop existing authority but cannot create or broaden it.

This stage prevents a previously valid authorization from overriding the world as it exists at execution time.

A.2.8 Execute: runtime coordination and domain actuation remain separate

Agent Runtime coordinates the workflow transition, governance call, provider invocation, timeout, retry, cancellation, budget, checkpoint and recovery behavior. It preserves one canonical execution state and fails closed for consequential transitions when governance is absent.

Cloud Scaling Operations performs the permitted Kubernetes or ArgoCD change. It receives external authorization and should execute only the operation that was authorized. Dry-run can be used during pilot and assurance testing before production mutation is enabled.

The separation matters: the controller recommends, authority components permit, the runtime coordinates and the operations package mutates infrastructure. No single component owns the entire chain.

A.2.9 Assure: successful dispatch is not assumed to be successful governance

Risk Authority Status Runtime continues to propagate revocation and authority-epoch changes. Runtime Assurance observes whether the workflow trajectory remains consistent with what governance approved. Execution Assurance reconciles the command, execution record and real-world effect.

For example, the Kubernetes API may report success while only 165 of the 180 requested replicas become ready. Latency may remain above the service objective because the bottleneck is a downstream database. Execution Assurance distinguishes command completion from intended effect and can trigger reassessment rather than allowing the system to assume success.

Context-Minimization Token-Accounting Runtime is relevant if LLM token consumption participates in workflow budgets. It is optional when the scaling decision path is entirely deterministic and uses no material model context.

A.2.10 Measure: readiness and value close the feedback loop

Agent Value Readiness can assess whether the governed scaling agent is ready for broader autonomy across intelligence fitness, capability readiness and organizational adoption. This is an advisory deployment input, not permission for the individual scaling action.

Governed Value can compare the authorized action's observed reliability benefit, avoided incident loss and incremental infrastructure cost. The result should distinguish modeled, reported and observed values. If the added capacity cost ₹400,000 while avoiding a verified outage exposure materially larger than that amount, the action may show positive governed value. If the predicted surge never occurred, the platform should not quietly report the forecast benefit as realized value.

These findings feed the next policy cycle: modify capacity limits, evidence requirements, readiness controls, escalation thresholds or model-selection policy based on observed outcomes rather than intuition alone.

A.3 Nine-stage walkthrough

Stage	Question answered	Scenario event	Primary output
Define	What rules and authority boundaries apply?	Issue scaling policy, compile Workflow IR and optionally bind the agent constitution.	Signed policy references and governed workflow
Propose	What should the AI recommend?	Recommend scaling the payments service from 120 to 180 replicas.	Identity-bound advisory and capacity recommendation
Verify	Can the claims, producer, evidence and policy be trusted?	Validate workload observations, producer identity and policy authenticity.	Evidence/control results and verification receipts
Decide	Has an accountable authority approved the business outcome?	Approve the bounded capacity increase for a limited period.	Binding decision and risk determination
Authorize	Does the authority permit this exact action?	Match actor, service, environment, operation and target replica count.	Exact-action authorization outcome
Clear	Is the action still safe now?	Recheck freeze, incident, current capacity and expiry immediately before mutation.	CLEAR, HOLD, ESCALATE or BLOCK
Execute	How is the permitted operation performed?	Coordinate and apply the exact Kubernetes or ArgoCD scaling change.	Canonical execution state and provider result
Assure	Did execution remain in bounds and produce the intended effect?	Reconcile requested, applied and ready replicas plus latency and errors.	Runtime/effect assessment and possible reassessment trigger
Measure	Was the system ready and was governed value created?	Evaluate autonomy readiness, reliability improvement and incremental cost.	Readiness determination and governed-value record

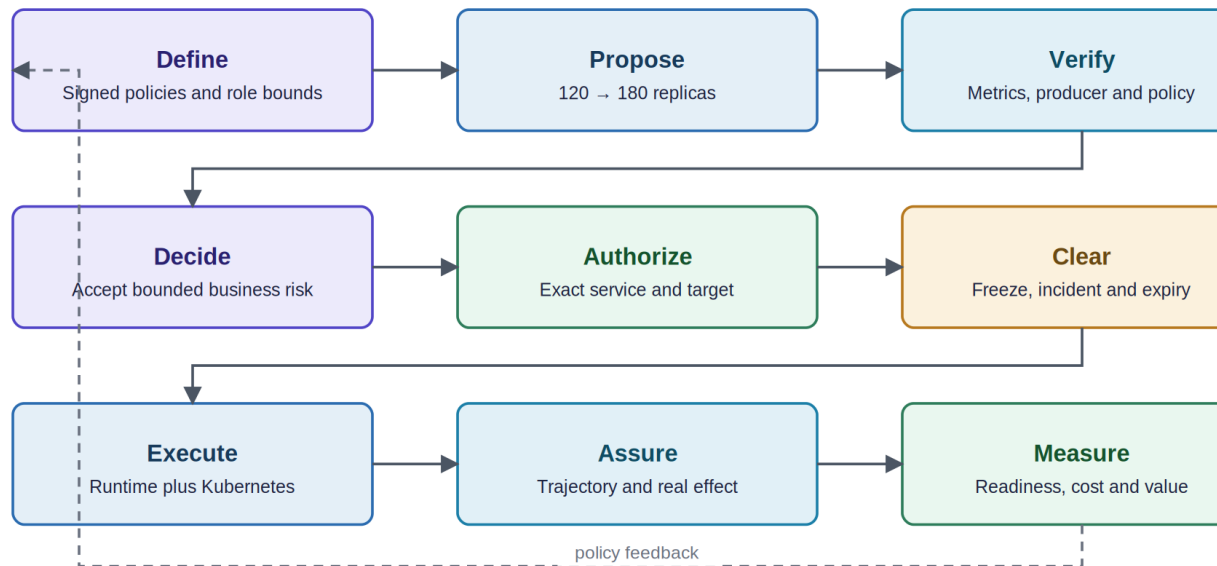


Figure 2. The nine stages applied to the governed cloud-scaling scenario.

A.4 Applicability classifications

Classification	Meaning in this appendix
Core required	Necessary for the high-assurance live governance path described here.
Domain required	Required because the example performs cloud-scaling analysis or mutation.

Classification	Meaning in this appendix
Optional enhancement	Adds useful control, intelligence or efficiency but can be omitted from the first bounded deployment.
Evaluation only	Used in design, testing, readiness or retrospective analysis rather than every live scaling transaction.
Not applicable	Unnecessary for the stated single-agent, fixed-execution scenario; it becomes relevant only if the scenario changes.

A.5 Complete 45-capability applicability matrix

#	Capability	Primary stage	Applicability	How it contributes—or why it is not needed
1	Governance Contracts	Foundation	Core required	Gives evidence, action, authority and execution components stable shared contracts.
2	Governance Provider Framework	Foundation	Core required	Composes TAP, ActionGate and provider boundaries without merging their authority.
3	JSON Canonicalization Scheme	Foundation	Core required	Creates deterministic identities for exact policy, proposal, evidence and action matching.
4	Benchmark Registry Contracts	Verify / Measure	Evaluation only	Useful when results are compared against governed benchmarks; not needed for a literal-threshold live decision.
5	UVI Policy Contracts	Define / Measure	Evaluation only	Supports governed readiness and value assessment rather than the immediate scaling mutation.
6	Policy Authority	Define	Core required	Issues, signs, resolves and revokes the exact scaling policy.
7	Policy Workflow Compiler	Define	Core required	Converts reviewed scaling policy into deterministic workflow and assurance artifacts.
8	Agent Constitution Policy	Define	Optional enhancement	Adds signed role bounds; an initial single-agent pilot may use externally enforced role restrictions.
9	Agent Constitution Activation	Define	Optional enhancement	Activates and maps the constitution when constitution governance is adopted.
10	Cloud Scaling Capacity-Bounds Policy	Define	Domain required	Expresses signed replica ceilings and scaling constraints.
11	Strategy-Permission Policy	Define / Propose	Optional enhancement	Governs declared reasoning strategies but grants no action authority.
12	Agent Constitution Conformance	Propose / Verify	Optional enhancement	Checks presented role facts against the signed constitution.
13	Agentic Proposer	Propose	Core required	Produces an inspectable identity-bound advisory rather than a direct command.

#	Capability	Primary stage	Applicability	How it contributes—or why it is not needed
14	Strategy-Permission Runtime Resolver	Propose / Verify	Optional enhancement	Resolves the exact signed strategy policy when strategy governance is enabled.
15	Agent Workforce Composer	Propose	Not applicable	One scaling agent is preassigned; required only for governed multi-agent team selection.
16	Reasoning Method Governance	Propose / Measure	Evaluation only	Supplies research contracts for method selection and comparison.
17	Reasoning Method Advisor	Propose	Evaluation only	Advises on reasoning methods during design, not on every live scaling event.
18	Readiness Comparison	Propose / Measure	Evaluation only	Compares method fit experimentally and produces no approval.
19	Trusted Workflow-Fit Pilot	Propose / Measure	Evaluation only	Tests reasoning workflows before production; current outputs remain research-only.
20	Model Selection	Propose	Optional enhancement	Needed only if more than one approved model/provider may be selected.
21	LLM Steering Controller	Propose	Optional enhancement	Adds dynamic routing, fallback and escalation when the workflow uses multiple LLM targets.
22	Context Minimization	Propose / Assure	Optional enhancement	Reduces sensitive context, latency and token use when an LLM participates.
23	StoryGraph	Propose / Verify	Optional enhancement	Detects sequence-level risk in a multi-step operational plan.
24	Cloud Scaling Controller	Propose	Domain required	Produces the explainable, non-executing scaling recommendation.
25	TAP	Verify	Core required	Evaluates whether claims behind the recommendation are supported by supplied evidence.
26	Trusted Evidence Authority	Verify	Core required	Establishes cryptographic evidence trust and independently verifiable receipts.
27	Benchmark Registry Authority	Verify / Measure	Evaluation only	Required for authoritative governed benchmark resolution when that future operational path is used; current package is not an authoritative live registry.
28	Risk Authority Evidence Runtime	Verify / Decide	Core required	Converts admitted evidence and control assurance into trusted Risk Authority inputs.

#	Capability	Primary stage	Applicability	How it contributes—or why it is not needed
29	Cloud Scaling Producer Attestation	Verify	Domain required	Proves the recommendation came from the expected producer.
30	Cloud Scaling Policy Authenticity	Verify	Domain required	Proves the candidate cites the exact current trusted capacity policy.
31	Decision Authority	Decide	Core required	Owens the binding business decision; the agent does not approve itself.
32	Risk Authority	Decide / Authorize	Core required	Applies mandatory controls and mints bounded, expiring and revocable machine authority.
33	Cloud Scaling Risk Integration	Decide	Domain required	Projects scaling facts into the enterprise risk case without executing them.
34	Cloud Scaling Authorization Contracts	Authorize	Domain required	Binds recommendation, risk, producer and references into one exact authorization candidate.
35	Risk Authority Runtime Composition	Authorize	Core required	Connects Decision Authority, Risk Authority and ActionGate fail-closed.
36	ActionGate	Authorize	Core required	Matches the exact actor, service, operation and target to supplied authority.
37	Action Clearance	Clear	Core required	Rechecks live freezes, incidents, state and expiry immediately before execution.
38	Agent Runtime	Execute	Core required	Coordinates the governed lifecycle, provider invocation, checkpoint and recovery.
39	Cloud Scaling Operations	Execute	Domain required	Applies the authorized Kubernetes or ArgoCD mutation.
40	Risk Authority Status Runtime	Assure	Core required	Propagates revocation and post-issuance authority status.
41	Risk Authority Runtime Assurance	Assure	Core required	Detects trajectory drift and can trigger authority reassessment.
42	Risk Authority Execution Assurance	Assure	Core required	Reconciles authorization, execution and actual effect.
43	Context-Minimization Token-Accounting Runtime	Assure / Measure	Optional enhancement	Settles measured LLM-token use when model calls participate in the workflow budget.
44	Agent Value Readiness	Measure	Evaluation only	Assesses whether the agent is ready for broader autonomy; it does not authorize this action.

#	Capability	Primary stage	Applicability	How it contributes—or why it is not needed
45	Governed Value	Measure	Optional enhancement	Connects observed reliability benefit and cost to the authorized action.

A.6 Minimum production path

The minimum path deliberately excludes research evaluation, dynamic model routing, multi-agent team composition and post-hoc value analytics. It contains only the governance and domain components required to move one consequential scaling proposal safely from policy to verified effect.



Figure 3. Minimum production path for one governed scaling action.

Required supporting foundations are Governance Contracts, Governance Provider Framework and deterministic canonical identity. The Policy Workflow Compiler and cloud-scaling integration contracts prepare the governed artifacts and domain bindings used by the path.

A.7 Capability-by-capability problem, solution and competitor analogue

A.7.1 Governance Contracts

The problem: Independent governance components can assign different meanings to the same action, evidence item or execution result, making composition and audit unreliable.

What it solves: Supplies neutral, reusable request/result and evidence/execution contracts without granting authority to the shared layer.

Competitor analogue: [Model Context Protocol](#) standardizes tool, resource and prompt contracts between LLM applications and servers, and [Agent2Agent](#) (Linux Foundation) standardizes agent discovery and task exchange. Both are open protocol standards rather than products, and neither defines evidence, decision, authority or execution-result contracts.

A.7.2 Governance Provider Framework

The problem: Evidence, authorization and execution providers need common discovery, lifecycle and failure mechanics, but a generic framework must not become the decision-maker.

What it solves: Provides provider registration, resolution, conformance and health mechanics while preserving each provider's authority boundary.

Competitor analogue: [Portkey AI Gateway](#) provides a partial analogue for provider routing, fallbacks and gateway controls; it is not equivalent to Ugence's separated evidence and action-authority contracts.

Additional analogues (cross-check supplement): [Cloudflare AI Gateway](#) fronts multiple model providers with rate limits, caching and cost budgets, a gateway-level analogue for provider composition rather than for evidence or action-governance provider contracts.

A.7.3 JSON Canonicalization Scheme

The problem: Equivalent JSON can serialize differently, breaking exact identity, signatures, comparison and replay.

What it solves: Produces deterministic canonical bytes and bare hashes so all stages can bind to the same exact artifact.

Competitor analogue: [Sigstore](#) signs artifact digests and records them in a tamper-evident log, and [OPA signed bundles](#) hash bundle files into a `.signatures.json`. Both depend on stable digests but neither is a canonicalization scheme for parsed JSON; no commercial product analogue was established.

A.7.4 Benchmark Registry Contracts

The problem: A benchmark can be silently changed, loosely referenced or applied outside its tenant, population, metric or time context.

What it solves: Makes benchmark identity exact and digest-bound while keeping trust and authoritative resolution separate.

Competitor analogue: [LangSmith Evaluation](#) provides datasets, experiment comparison and evaluation workflows, but not the same digest-bound benchmark-authority model.

Additional analogues (cross-check supplement): [Stanford HELM](#) standardizes scenarios and metrics across models, and [W&B Registry](#) assigns immutable versions and lineage to artifacts; neither binds a benchmark identity to tenant, population, metric and period coordinates.

A.7.5 UVI Policy Contracts

The problem: Readiness and value claims become incomparable or gameable when geography, intended outcome, evidence requirements and thresholds are implicit.

What it solves: Provides immutable governed context and policy shapes for later readiness and value engines.

Competitor analogue: [IBM watsonx.governance](#) partially overlaps through AI lifecycle, risk, control and accountability records.

Additional analogues (cross-check supplement): [OneTrust AI Governance](#) and [Holistic AI](#) inventory AI systems and assess them against frameworks such as the NIST AI RMF and the EU AI Act, partial overlaps for assessment context rather than immutable value-policy shapes.

A.7.6 Policy Authority

The problem: Editable policies and local configuration cannot prove which approved version governed an action or whether it was revoked.

What it solves: Issues, signs, registers, resolves and revokes exact policy artifacts shared across the platform.

Competitor analogue: [Credo AI Policy Packs](#) translate governance requirements into reusable policy controls; [Open Policy Agent](#) provides policy-as-code evaluation. Neither source establishes the same signed multi-family issuance and revocation architecture described here.

Additional analogues (cross-check supplement): [Amazon Verified Permissions](#) centralizes Cedar policy management with versioned policy stores, and [Cerbos](#) manages policies through a policy administration plane; both are partial analogues for issuance and resolution without the signed multi-family revocation model.

A.7.7 Policy Workflow Compiler

The problem: Runtime interpretation of prose policies is nondeterministic and difficult to test before deployment.

What it solves: Compiles reviewed structured policy into deterministic Workflow IR, assurance specifications and content-addressed artifacts.

Competitor analogue: [Credo AI](#) advertises policy-to-code translation, automated workflows and audit-ready evidence; [OPA's Rego](#) expresses policy decisions over structured data. These are partial rather than exact compiler analogues.

A.7.8 Agent Constitution Policy

The problem: Agent roles can expand their declared actions, dispositions or tool scopes when boundaries live only in prompts or mutable configuration.

What it solves: Makes the role's structural ceiling an issuable, signed, versioned and revocable policy artifact.

Competitor analogue: [Amazon Bedrock AgentCore Identity](#) and [AgentCore Policy](#) partially overlap through agent identity and externalized fine-grained permissions, but do not document the same signed constitution vocabulary.

Additional analogues (cross-check supplement): [Auth0 for AI Agents](#) (Okta) adds agent authentication, fine-grained authorization and asynchronous approval to the identity layer, a partial analogue for role bounds expressed as identity permissions.

A.7.9 Agent Constitution Activation

The problem: An issued constitution is not operationally useful until its exact references and composition dependencies are activated safely.

What it solves: Performs preflight, derives governed reference mappings and records key-material-free activation receipts.

Competitor analogue: [Microsoft Entra Agent ID](#) provisions agent identities from blueprints with lifecycle governance, and [Workday Agent System of Record](#) registers, configures, activates and deactivates agents. Both are partial analogues for the activation lifecycle; neither derives governed reference maps from a signed constitution or issues key-material-free activation receipts.

A.7.10 Cloud Scaling Capacity-Bounds Policy

The problem: A controller can otherwise treat replica limits as adjustable tuning rather than organization-approved ceilings.

What it solves: Makes capacity ceilings signed, versioned and resolvable through Policy Authority.

Competitor analogue: [Sedai Kubernetes Optimization](#) describes safe optimization against workload behavior and SLOs, while [Open Policy Agent](#) can express infrastructure constraints. The Ugence module specifically binds capacity ceilings to its policy-authority path.

Additional analogues (cross-check supplement): [Kyverno](#) enforces Kubernetes admission policies, including signed-artifact checks, a partial analogue for cluster-side limits expressed as policy.

A.7.11 Strategy-Permission Policy

The problem: An agent may select an inappropriate reasoning procedure without an organization-governed permission boundary.

What it solves: Defines the signed set of reasoning strategies a governed role may declare, without authorizing tools, compute or action.

Competitor analogue: [NVIDIA NeMo Guardrails](#) constrains dialog flows and actions through Colang rails, and [DSPy](#) makes reasoning modules such as chain-of-thought and ReAct explicit program choices. Neither expresses a signed, versioned and revocable permission over which reasoning strategies a governed role may declare.

A.7.12 Agent Constitution Conformance

The problem: Issuing a constitution does not prove that the current presented role remains inside its signed bounds.

What it solves: Resolves the exact constitution and performs a fail-closed subset check over governed role facts.

Competitor analogue: [AgentCore Policy](#) externally checks fine-grained permissions for identity and tool-input parameters, a partial runtime-control analogue rather than a constitution-conformance verifier.

A.7.13 Agentic Proposer

The problem: Free-form agent output is difficult to govern and may be mistaken for a decision or command.

What it solves: Produces typed, identity-bound candidates and advisories with explicit evidence needs, abstention and escalation states.

Competitor analogue: [HumanLayer](#) gates tool calls behind human approval routed over Slack or email, and [LangGraph human-in-the-loop](#) interrupts let a reviewer approve, edit or reject a proposed tool call. Both implement propose-then-approve at the tool-call layer without typed candidates, evidence requests, abstention states or identity-bound advisories.

A.7.14 Strategy-Permission Runtime Resolver

The problem: A proposer cannot safely trust a local list of permitted reasoning strategies or resolve policy by an unsigned floating reference.

What it solves: Resolves the exact signed policy through Policy Authority and fails closed on tenant, scope, lifecycle, approval or reference mismatch.

Competitor analogue: [Cerbos](#) and [Amazon Verified Permissions](#) (Cedar) resolve externalized policy at runtime for agent actions. They are partial analogues for runtime policy resolution, but they answer allow or deny on an action rather than resolve a signed set of permitted reasoning strategies with tenant, scope and lifecycle checks.

A.7.15 Agent Workforce Composer

The problem: Multi-agent workflows can assign roles based on availability or model preference without hard capability, evidence and least-privilege constraints.

What it solves: Determines eligibility, ranks qualified agents, composes bounded teams and proposes least-privilege permissions and fallbacks.

Competitor analogue: [CrewAI](#) composes role-based agent crews, [Microsoft Agent Framework](#) supplies sequential, concurrent, handoff and group-chat orchestration patterns, and [Workday Agent System of Record](#) registers and governs an agent workforce. None documents hard-constraint eligibility with complete elimination accounting or least-privilege permission proposals.

A.7.16 Reasoning Method Governance

The problem: Reasoning-method experiments use inconsistent task, telemetry, evidence and fit vocabularies.

What it solves: Supplies shared research-only contracts so advice, comparison and pilot evidence remain distinguishable and interoperable.

Competitor analogue: [LangSmith Evaluation](#) provides a framework for measuring agent quality from predeployment testing through production monitoring, a partial analogue for evaluation vocabulary and workflow.

Additional analogues (cross-check supplement): [Braintrust](#) records experiments as immutable, comparable eval runs, a partial analogue for the evidence and comparison vocabulary.

A.7.17 Reasoning Method Advisor

The problem: Developers may choose reasoning methods by fashion, intuition or an LLM's unexamined preference.

What it solves: Deterministically identifies zero, one or many qualifying methods from governed task characteristics and explains every inclusion and exclusion.

Competitor analogue: [DSPy](#) optimizers search over instructions and demonstrations for a declared module, and [Braintrust experiments](#) compare configurations empirically. Both are optimization or evaluation tools, not a deterministic rule-derived advisor that explains every inclusion and exclusion and refuses to manufacture a winner.

A.7.18 Readiness Comparison

The problem: Rule-derived advice alone does not establish that one reasoning method performs better for the task.

What it solves: Provides a deterministic research comparison over declared quality and resource dimensions without manufacturing authority.

Competitor analogue: [LangSmith Evaluation](#) supports experiment comparison, datasets and regression evaluation, but does not document Ugence's authority-separated readiness contract.

Additional analogues (cross-check supplement): [Braintrust experiments](#) and [Datadog LLM Observability](#) evaluations compare configurations on quality and operational metrics without an authority-separated readiness contract.

A.7.19 Trusted Workflow-Fit Pilot

The problem: Workflow-method evaluations can be biased by post-hoc case selection, self-reported telemetry or undeclared evaluator behavior.

What it solves: Preregisters the study, isolates capture, recomputes telemetry and preserves research-only evidence lineage.

Competitor analogue: [LangSmith's CI/CD evaluation pattern](#) shows agent evaluation in deployment pipelines; the Ugence pilot adds its own preregistration and authority-label separation.

A.7.20 Model Selection

The problem: Agents can route work to models that are unapproved, incompatible with policy or unsuitable for the task.

What it solves: Deterministically selects only from the approved eligible model/provider set without performing routing or execution.

Competitor analogue: [Portkey AI Gateway](#) supports a model catalog, conditional routing and provider fallbacks, partially overlapping selection and routing.

Additional analogues (cross-check supplement): [Not Diamond](#), [Amazon Bedrock Intelligent Prompt Routing](#) and the [Microsoft Foundry model router](#) select a model per request, with the Foundry router supporting model subsets that restrict routing to approved models. All three combine selection with routing and execution, which the Ugence capability deliberately separates.

A.7.21 LLM Steering Controller

The problem: Static model routing creates cost, latency, reliability and policy failures when conditions change.

What it solves: Produces explainable provider-neutral routing, fallback and escalation recommendations after hard-constraint filtering.

Competitor analogue: [Portkey Conditional Routing](#) routes requests to provider targets under custom conditions and supports fallbacks through its gateway.

Additional analogues (cross-check supplement): [Not Diamond](#), [Amazon Bedrock Intelligent Prompt Routing](#) and the [Microsoft Foundry model router](#) are learned or managed routers that also execute the call; the Ugence controller produces only an advisory recommendation.

A.7.22 Context Minimization

The problem: Agents often receive more sensitive, costly and distracting context than the task requires.

What it solves: Removes unnecessary context while protecting required units and preserving a declared equivalence condition, with measurable token accounting.

Competitor analogue: [LLMLingua](#) (Microsoft Research) compresses prompts with a small language model, and [Private AI](#) and [Presidio](#) detect and redact personal data before LLM calls. These are partial analogues for token reduction and sensitive-data removal; none preserves a caller-declared equivalence condition with fail-closed refusal.

A.7.23 StoryGraph

The problem: A sequence of individually permitted actions can create a collectively prohibited or unsafe outcome.

What it solves: Evaluates sequence-level risk and emits advisory evidence before binding authorization.

Competitor analogue: [Amazon Bedrock AgentCore's multi-action policy controls](#) describe controls over agent behavior and cost beyond one stateless action, a partial analogue to sequence-aware control.

Additional analogues (cross-check supplement): [Lakera Guard](#) detects direct and indirect prompt injection across agent interactions, and [Zenity](#) and [Noma Security](#) monitor and enforce on agent tool calls at runtime; these overlap threat detection on behavior sequences rather than policy-defined sequence-risk evidence.

A.7.24 Cloud Scaling Controller

The problem: Reactive manual scaling is slow, while an autonomous optimizer that also executes creates excessive authority concentration.

What it solves: Generates deterministic, explainable scaling advice with no mutation capability.

Competitor analogue: [Sedai](#) provides autonomous cloud and Kubernetes optimization; [CAST AI](#) provides Kubernetes cost optimization and autoscaling. Both overlap the operational domain, while Ugence deliberately separates recommendation from authorization and execution.

Additional analogues (cross-check supplement): [StormForge Optimize Live](#) and [PerfectScale by DoiT](#) rightsizes Kubernetes workloads autonomously, and [Karpenter](#) provisions nodes in response to unscheduled pods; all three recommend and act inside one component.

A.7.25 TAP

The problem: Confident model assertions and incomplete evidence can be accepted as facts by downstream decision systems.

What it solves: Classifies whether an assertion is supported, unsupported, constrained or indeterminate without granting action authority.

Competitor analogue: [Portkey Guardrails](#) can verify LLM inputs and outputs against configured checks; [Credo AI](#) records governance evidence. These partially overlap but do not document TAP's separate assertion-governance outcome model.

Additional analogues (cross-check supplement): [Vectara HHEM](#), [Patronus Lynx](#) and [Cleanlab TLM](#) score whether a response is supported by its context or is trustworthy, and [Guardrails AI](#) packages such checks as validators. These produce probabilistic scores rather than typed supported, unsupported, constrained or indeterminate outcomes with a fail-safe unknown state.

A.7.26 Trusted Evidence Authority

The problem: A signature, evidence reference or claimed provenance is not enough to establish current trust.

What it solves: Verifies evidence identity, trust anchors, key entitlement, revocation, scope and signing frames and issues independently re-verifiable receipts.

Competitor analogue: [Sigstore](#) (cosign and Rekor) signs artifacts and attestations and records them in a transparency log, and [SPIFFE/SPIRE](#) issues short-lived, rotated workload identities. Both are partial analogues for signature trust, key validity and provenance; neither issues an evidence-verification receipt bound to scope and revocation of the evidence itself.

A.7.27 Benchmark Registry Authority

The problem: Benchmark retrieval can be mistaken for authoritative admission, approval and exact resolution.

What it solves: Defines the lifecycle and trust boundary for future authoritative benchmark handling while honestly withholding capabilities not yet implemented.

Competitor analogue: [MLflow Model Registry](#) and [W&B Registry](#) version and govern artifacts with lineage, and [Stanford HELM](#) publishes standardized benchmark leaderboards. These are partial analogues for versioned registries and benchmark publication; none establishes digest-bound benchmark admission, trusted resolution and revocation as an authority.

A.7.28 Risk Authority Evidence Runtime

The problem: Risk decisions cannot safely consume caller-asserted evidence status or disconnected control results.

What it solves: Composes evidence admission and assertion assurance into trusted, non-compensatory control inputs.

Competitor analogue: [Credo AI](#) combines policy controls, automated workflows and audit-ready evidence; this is a partial governance-process overlap rather than an equivalent runtime authority input.

A.7.29 Cloud Scaling Producer Attestation

The problem: A valid-looking scaling candidate may have been created by an unknown, replaced or unauthorized producer.

What it solves: Binds verified producer authenticity to the exact scaling authorization candidate without granting permission.

Competitor analogue: [Amazon Bedrock AgentCore Identity](#) provides agent identity and controlled resource access, a partial producer-identity analogue.

Additional analogues (cross-check supplement): [Sigstore](#) attestations bind a producer identity to an artifact through a transparency log, and [SPIFFE/SPIRE](#) attests workload identity at runtime; both are close analogues for producer authenticity without binding to an authorization candidate.

A.7.30 Cloud Scaling Policy Authenticity

The problem: A recommendation can cite an expired, revoked, mismatched or fabricated capacity policy.

What it solves: Verifies that the referenced policy is the exact trusted Policy Authority artifact applicable to the candidate.

Competitor analogue: [OPA signed bundles](#) activate a policy only after signature verification against a configured key, and [Kyverno](#) verifies Sigstore signatures on images and OCI artifacts. Both are close analogues for policy-artifact authenticity; neither reconciles tenant, scope, coordinate and time window against a specific authorization candidate.

A.7.31 Decision Authority

The problem: Recommendations, risk scores and evidence results are often allowed to become de facto business approvals without a clearly accountable owner.

What it solves: Owns the bounded binding business decision and its delegation, scope, conditions and validity.

Competitor analogue: [IBM watsonx.governance Model Risk Governance](#) documents staged model-governance workflows through deployment approval, a partial governance-approval analogue.

Additional analogues (cross-check supplement): [ServiceNow AI Control Tower](#) assigns human managers to oversee agents, and [OneTrust AI Governance](#) automates approval and attestation workflows; both are partial analogues for accountable approval without a bounded, scoped, time-limited binding decision record.

A.7.32 Risk Authority

The problem: Human approval is not directly machine-enforceable and can be broadened or reused after conditions change.

What it solves: Converts an approved decision into scoped, time-bound, revocable and cryptographically bound machine authority after mandatory controls pass.

Competitor analogue: [AgentCore Policy](#) provides fine-grained permissions based on identity and tool-input parameters. It overlaps runtime policy enforcement but does not document the same decision-derived authorization-envelope chain.

Additional analogues (cross-check supplement): [Holistic AI](#) and [OneTrust AI Governance](#) assess AI risk against regulatory frameworks, and [Amazon Verified Permissions](#) and [Cerbos](#) enforce machine-readable authorization; none converts a risk determination into a signed, expiring, revocable authority envelope.

A.7.33 Cloud Scaling Risk Integration

The problem: Domain controller outputs do not naturally fit a general enterprise risk case.

What it solves: Projects scaling observations and recommendations into Risk Authority's governed risk representation without importing authority into the controller.

Competitor analogue: [IBM watsonx.governance](#) and [Credo AI](#) provide broader AI risk and control workflows, but neither cited source establishes this exact scaling-to-machine-authority adapter.

A.7.34 Cloud Scaling Authorization Contracts

The problem: Authorization can drift if the recommendation, producer, risk result and policy references are reconstructed separately.

What it solves: Binds them into one exact, digestible capacity-action candidate that explicitly grants nothing.

Competitor analogue: [HCP Terraform run tasks](#) gate a specific plan before apply with an external pass or fail, and [Argo Rollouts analysis](#) binds promotion of a specific rollout step to metric results. Both are partial analogues for binding one exact change to an approval, without a candidate object that unifies recommendation, risk result, producer identity and policy references.

A.7.35 Risk Authority Runtime Composition

The problem: Separate decision, risk and action components can be wired inconsistently or bypassed by application code.

What it solves: Provides a fail-closed composition of Decision Authority, Risk Authority and ActionGate.

Competitor analogue: [Amazon Bedrock AgentCore](#) combines identity, gateway and policy controls for agent tool access, a partial platform-level analogue; Ugence additionally separates the binding business decision and machine-authority artifact.

A.7.36 ActionGate

The problem: A legitimate general approval can be replayed for a different actor, tool, resource, input or operation.

What it solves: Evaluates whether the exact proposed action matches supplied authority and context without executing it.

Competitor analogue: [AgentCore Policy](#) intercepts agent tool calls and evaluates fine-grained permissions outside the reasoning loop; [OPA](#) provides general action/resource authorization policy. These are the closest documented overlaps.

Additional analogues (cross-check supplement): [Cerbos](#) and [Amazon Verified Permissions](#) evaluate principal, action, resource and context per request, and [Zenity](#) enforces on agent actions at decision time; these are the closest commercial overlaps for exact-action checks.

A.7.37 Action Clearance

The problem: Conditions can change after authorization but before execution.

What it solves: Rechecks trusted current-state signals and may hold, escalate or block without creating or broadening authority.

Competitor analogue: [OPA external-data guidance](#) supports context-aware policy decisions using current external state, a partial analogue; the cited source does not describe Ugence's separate post-authorization clearance stage.

Additional analogues (cross-check supplement): [ServiceNow Change Management blackout and maintenance schedules](#) encode freeze windows and detect scheduling conflicts, a partial analogue for the freeze and conflict signals Action Clearance consumes.

A.7.38 Agent Runtime

The problem: Agent workflows need reliable lifecycle, retries, cancellation, budgets, concurrency, checkpointing and recovery without allowing orchestration to become governance authority.

What it solves: Coordinates governed execution and maintains canonical execution state while delegating consequential authorization to an external boundary.

Competitor analogue: [Amazon Bedrock AgentCore](#) provides managed agent runtime, identity, gateway and observability capabilities; the Ugence package is a domain-neutral kernel with an explicitly external governance boundary.

Additional analogues (cross-check supplement): [Temporal](#) provides durable execution with retries, timeouts and replay for agent workflows, [LangGraph](#) adds checkpointing and interrupts, and [Microsoft Agent Framework](#) and [CrewAI](#) supply orchestration runtimes; none documents an external fail-closed governance boundary on consequential transitions.

A.7.39 Cloud Scaling Operations

The problem: A recommendation requires controlled domain actuation, but embedding credentials and mutation inside the recommender defeats governance separation.

What it solves: Executes externally authorized Kubernetes or ArgoCD changes with a dry-run-first posture.

Competitor analogue: [Sedai Kubernetes Optimization](#) performs autonomous production optimization; [CAST AI](#) automates Kubernetes optimization and scaling.

Additional analogues (cross-check supplement): [Argo Rollouts](#) performs progressive Kubernetes delivery with metric-gated promotion, and [Karpenter](#) actuates node capacity; both mutate infrastructure under their own control loops rather than under an external authorization.

A.7.40 Risk Authority Status Runtime

The problem: Once issued, machine authority can remain usable after revocation or policy/risk changes unless status propagates.

What it solves: Manages post-issuance revocation and authority-epoch propagation.

Competitor analogue: [AgentCore Identity](#) provides agent access control and identity infrastructure, a partial lifecycle analogue; the cited material does not establish the same authorization-epoch design.

Additional analogues (cross-check supplement): [SPIFFE/SPIRE](#) bounds identity lifetime through short-lived, automatically rotated credentials, and [Auth0 Token Vault](#) manages token refresh and exchange for agents; both are partial analogues for post-issuance lifecycle without authority-epoch propagation.

A.7.41 Risk Authority Runtime Assurance

The problem: A workflow can diverge from the authorized trajectory even when individual provider calls succeed.

What it solves: Observes trajectory and routes material drift back into authority reassessment without minting authority itself.

Competitor analogue: [LangSmith Observability](#) records agent traces for debugging, quality monitoring and evaluation; [AgentCore Observability](#) emits runtime traces and metrics. These overlap observation, not the full authority-reassessment path.

Additional analogues (cross-check supplement): [Datadog LLM Observability](#) traces and evaluates agent runs, and [Noma Security](#), [Zenity](#) and [ServiceNow AI Control Tower](#) detect off-policy agent behavior and can stop an agent; these overlap observation and interruption, not routing drift back into authority reassessment.

A.7.42 Risk Authority Execution Assurance

The problem: Dispatch success does not prove that the authorized real-world effect occurred or remained within bounds.

What it solves: Reconciles authorization, execution evidence and observed effect and can trigger a new risk/decision assessment.

Competitor analogue: [Sedai Kubernetes Optimization](#) tracks workload behavior, application performance, cost drivers and SLO impact, a partial operational-effect analogue; Ugence binds effect reconciliation back to authority.

Additional analogues (cross-check supplement): [Harness Continuous Verification](#) verifies a deployment against metrics and logs and can roll back automatically, and [Argo Rollouts analysis](#) aborts a rollout on failed metric queries; both are close analogues for reconciling intended and observed effect in the delivery domain.

A.7.43 Context-Minimization Token-Accounting Runtime

The problem: Estimated token use can understate actual provider consumption and weaken shared budget enforcement.

What it solves: Converts provider-attempt telemetry into reconciled context/token records and settles budgets from measured use.

Competitor analogue: [Portkey's agent gateway controls](#) log cost, tokens, latency and actor and support budget/rate limits, a close operational accounting overlap.

Additional analogues (cross-check supplement): [Helicone](#) and [Langfuse](#) attribute token usage and cost per request and span with budget alerts, and [Cloudflare AI Gateway](#) enforces cost-based budgets at the gateway; all are close operational analogues for measured accounting.

A.7.44 Agent Value Readiness

The problem: Organizations can confuse technical capability or a successful demo with readiness for an intended enterprise outcome.

What it solves: Produces a governed, non-financial and multidimensional readiness determination covering intelligence, capability and adoption.

Competitor analogue: [IBM watsonx.governance](#) tracks model facts, lifecycle performance and risk management, a partial readiness-governance overlap.

Additional analogues (cross-check supplement): [ServiceNow AI Control Tower](#) and [Holistic AI](#) track AI assets, risk posture and governance status, partial analogues for adoption and capability readiness without a deterministic multidimensional determination.

A.7.45 Governed Value

The problem: Forecast savings, reported benefits and observed realized value are often collapsed into one optimistic ROI figure.

What it solves: Attributes net governed value to authorized actions while preserving evidence, authority and outcome classifications.

Competitor analogue: [Sedai Cloud Cost Optimization](#) ties cloud savings to optimization actions and audit receipts, a close domain overlap; Ugence's capability is intended as a cross-domain governance accounting kernel.

Additional analogues (cross-check supplement): [ServiceNow AI Control Tower](#) consolidates ROI, productivity, cost avoidance and risk-reduction metrics per AI asset, and [Workday Agent System of Record](#) measures agents like investments; both report value without separating reported, modeled and observed evidence classes.

A.8 Essential deployment stack versus later enhancements

A.8.1 Essential platform controls

The high-assurance production path requires:

- Governance Contracts, Governance Provider Framework and canonical identity;
- Policy Authority and the Policy Workflow Compiler;
- Agentic Proposer;
- TAP and Trusted Evidence Authority;
- Risk Authority Evidence Runtime;
- Decision Authority and Risk Authority;
- Risk Authority Runtime Composition, ActionGate and Action Clearance;
- Agent Runtime; and
- Risk Authority Status, Runtime Assurance and Execution Assurance.

These capabilities implement the platform's non-collapsible accountability chain. Removing one requires an explicit alternative control; otherwise the system risks confusing policy, recommendation, evidence, decision, authorization, execution or observed effect.

A.8.2 Essential cloud-scaling adapters and operations

The scenario additionally requires:

- Cloud Scaling Controller;
- Cloud Scaling Capacity-Bounds Policy;
- Cloud Scaling Risk Integration;
- Cloud Scaling Authorization Contracts;
- Cloud Scaling Producer Attestation;
- Cloud Scaling Policy Authenticity; and
- Cloud Scaling Operations.

These do not redefine enterprise governance. They translate the general platform into the language of replicas, workload observations, capacity limits and Kubernetes or ArgoCD operations.

A.8.3 Optional production enhancements

The following deepen control or operational efficiency but need not block the first bounded single-agent deployment:

- Agent Constitution Policy, Activation and Conformance;
- Strategy-Permission Policy and Runtime Resolver;
- Model Selection and LLM Steering Controller;
- Context Minimization and its token-accounting integration;
- StoryGraph; and
- Governed Value.

They should be enabled when the deployment introduces dynamic model choice, sensitive context, complex multi-step behavior, reusable agent roles, token budgets or a formal value-realization program.

A.8.4 Evaluation and future-governance capabilities

The following should operate outside the per-action critical path:

- Benchmark Registry Contracts and the present Benchmark Registry Authority surface;
- UVI Policy Contracts and Agent Value Readiness;
- Reasoning Method Governance, Reasoning Method Advisor and Readiness Comparison; and
- Trusted Workflow-Fit Pilot.

They support research, benchmark governance, readiness and system improvement. Their evidence can influence later policies and deployment decisions, but research-only or contract-only outputs must not be promoted into live action authority.

A.8.5 Not required in the stated minimum scenario

Agent Workforce Composer is not required because the scenario uses one preassigned scaling agent. It becomes valuable if the enterprise later composes forecasting, finance, reliability and operations agents dynamically under capability, evidence and least-privilege constraints.

A.9 Competitive interpretation

The cited products demonstrate that parts of the Ugence architecture have market analogues:

- Credo AI and IBM watsonx.governance overlap policy, risk, lifecycle and evidence-management functions.
- Amazon Bedrock AgentCore overlaps agent identity, tool-call policy enforcement, runtime and observability.
- Open Policy Agent overlaps policy-as-code and context-aware authorization.
- Portkey overlaps model routing, gateway guardrails, observability and token/cost controls.
- LangSmith overlaps agent tracing, evaluation, datasets and deployment-pipeline testing.
- Sedai and CAST AI overlap autonomous cloud or Kubernetes optimization and execution.
- StormForge, PerfectScale and Karpenter add further Kubernetes rightsizing and autoscaling overlap, again combining recommendation with actuation.
- Microsoft Entra Agent ID, Okta/Auth0 for AI agents, SPIFFE/SPIRE and Workday Agent System of Record overlap agent identity, provisioning, activation and credential lifecycle.
- Zenity, Noma Security and Lakera overlap runtime agent security, tool-call enforcement and injection detection.
- Vectara HHEM, Patronus Lynx, Cleanlab TLM and Guardrails AI overlap assertion and faithfulness scoring in the assessment path.
- Not Diamond, Amazon Bedrock Intelligent Prompt Routing and the Microsoft Foundry model router overlap model selection and routing.
- Cloudflare AI Gateway, Helicone, Langfuse, Datadog LLM Observability and Braintrust overlap gateway controls, token accounting, tracing and evaluation.
- Sigstore, OPA signed bundles and Kyverno overlap artifact and policy authenticity; Harness Continuous Verification, Argo Rollouts and HCP Terraform run tasks overlap change gating and effect verification.
- ServiceNow AI Control Tower, OneTrust AI Governance and Holistic AI overlap AI inventory, approval workflow, risk assessment and value reporting.
- Temporal, LangGraph, Microsoft Agent Framework, CrewAI and HumanLayer overlap agent runtime, orchestration and human approval.
- Model Context Protocol and Agent2Agent overlap contract standardization for tools and agent interoperability.

The comparison should not claim that a blank competitor field proves uniqueness. It means only that no sufficiently close analogue was established from the limited official-source review used for this appendix, as supplemented by the cross-check in Appendix C. The more defensible Ugence distinction is architectural: the repository separates proposal, assertion verification, binding business decision, risk-derived machine authority, exact-action authorization, operational clearance, execution coordination, runtime assurance, effect reconciliation and governed-value measurement into explicit non-collapsible responsibilities.

Appendix B — Development Status of the 45 Capabilities

B.1 Purpose and evidence basis

This appendix answers one question per capability: **where does development stand today?** Each row is derived from the package as it exists under `packages/` in the inspected snapshot: the distribution version, the package README's own status, maturity, phase and scope statements, and the size of its test suite. It does not restate architectural role (Sections 4–13) or scenario applicability (Appendix A).

Evidence labels follow the repository working agreement: `[V]` verified against the package itself, `[I]` inferred from adjacent repository material. Unless a row is marked `[I]`, every statement in it is `[V]` from the package's README and metadata.

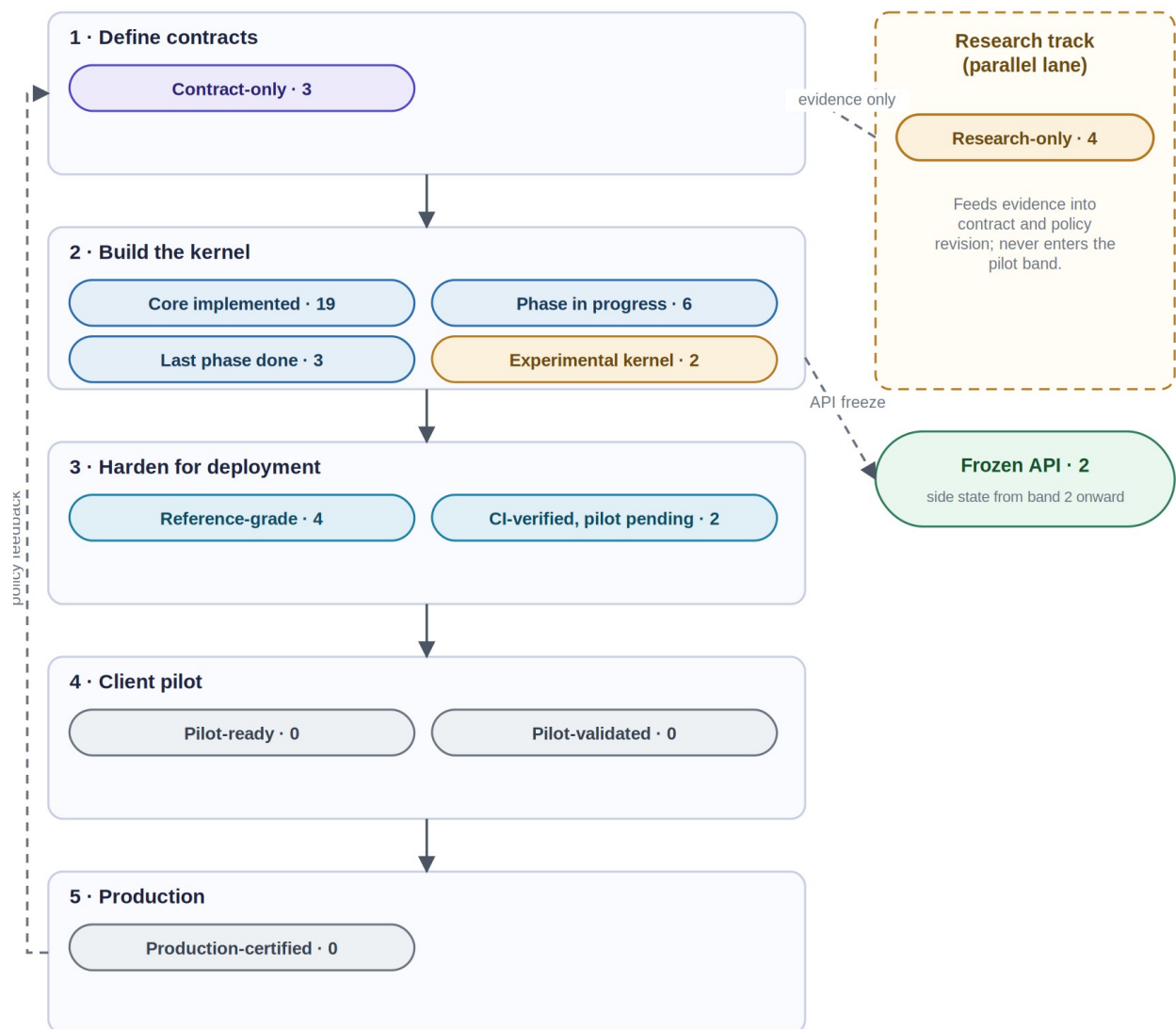
The single most important finding is stated first: **no package among the 45 declares itself pilot-validated, shadow-deployed or production-certified.** Many READMEs disclaim it explicitly; the rest simply make no such claim. The closest pilot evidence in the repository sits outside these packages: the legacy Enterprise Validation Pilot (Phase 5I, `docs/ENTERPRISE_VALIDATION_PILOT.md`) exercised the predecessor distributions of Decision Authority, TAP and ActionGate together, and the excluded AI Hiring product declares `PACKAGE_READY_FOR_CONTROLLED_PILOT`. Both are noted as `[I]` where relevant.

B.2 Stage tags

Tag	Meaning
Contract-only	Typed contracts, vocabulary or validation only; no operational engine, store, verifier or clock.
Research-only	Experimental study path whose every output is labelled non-approval-bearing by construction.
Experimental kernel	A working calculation or determination kernel that the package itself marks experimental and advisory.
Core implemented	Deterministic kernel implemented and unit-tested; no phase ladder pending inside the package; no pilot or production claim.
Phase in progress	Part of a numbered phase ladder whose later phases are named and still pending.
Last phase done	The package's own phase sequence is complete; any remaining work is assigned to a different package or milestone.
Reference-grade	Operative logic shipped with in-memory or reference adapters that are refused when <code>production_mode</code> is set; production adapters delegated.
CI-verified, pilot pending	Implemented and CI-verified, and the README itself states that pilot or production validation remains pending.
Frozen	Public API frozen at a major version; no phase ladder; validation described as synthetic or legacy-lineage.
Pilot-ready	The package declares readiness to start a bounded or controlled pilot with a client. No capability currently carries this tag.
Pilot-validated	A pilot has run against this package lineage and its results are recorded. No capability carries it; several READMEs state the negative (<code>pilot_validated=false</code> , "Not pilot-validated").
Production-certified	Formal certification beyond pilot. No capability carries it; every README that mentions it states the negative.

B.3 Canonical development pipeline

The stage tags are not an arbitrary list. They sit at fixed points on one development cycle that every capability is expected to travel: contracts first, then an implemented kernel, then hardening, then a client pilot, then production. The research track runs beside that cycle and feeds evidence back into definition; it never enters the pilot band on its own. **Frozen** is an API-stability state that can be reached at any point from the kernel band onward, so it is drawn as a side state rather than a band.



All 45 capabilities sit in bands 1–3. Bands 4 and 5 are empty by each package's own declaration.

Figure 4. Canonical development pipeline with the stage tags placed on it.

How to read it against the table in B.4:

- **Bands 1–3 hold all 45 capabilities.** Nothing has crossed into band 4, which is the finding stated in B.1.
- **Band 2 is where a package's own phase ladder lives.** "Phase in progress" and "Last phase done" describe position on that ladder; "Core implemented" means the package has no ladder left inside it. The cloud-scaling thread is the clearest example of a ladder that spans several packages, with Phases 1–5B landed and 5B-2, 5C, 5X, 5D and 6 still unbuilt.
- **Band 3 is the step the Risk Authority runtimes and Agent Runtime have reached.** Reference-grade means the logic is operative but its production adapters are delegated; CI-verified, pilot pending means the README itself names pilot or production validation as the next step.

- **Band 4 has two rungs.** Pilot-ready means the package declares fitness to start a controlled pilot with a client. Pilot-validated means a pilot has run and its results are recorded. The legacy Enterprise Validation Pilot sits near the second rung but ran over predecessor distributions, so it is cited as inferred and not counted.
- **The research lane is not a shortcut.** Research-only and Experimental-kernel outputs reach the main cycle only as evidence for policy and contract revision, never as authority to deploy.

B.4 Status table

Test counts are def `test_` occurrences under each package's `tests/` tree at the inspected commit; they indicate suite size, not coverage or pass state.

#	Capability	Version	Stage tag	Phase position	Where development stands	Tests
1	Governance Contracts	0.3.1	Core implemented	Extraction, GV-2E-a, M-3R.3 and 0.3.1 canonicalization done; contract-evolution phase pending	Neutral contracts extracted verbatim from the frozen core; authenticity fields stay permanently <code>STRUCTURAL_UNVERIFIED</code> because no ratified system-binding verifier exists yet.	104
2	Governance Provider Framework	0.1.0	Core implemented	Contract version 1.0.0; no phase ladder	Provider registry, resolution and conformance mechanics are in place with reference providers for framework validation only; the README makes no maturity statement. Predecessor distribution was exercised by the legacy Enterprise Validation Pilot [I].	57
3	JSON Canonicalization Scheme	0.2.0	Core implemented	Alpha; extracted from <code>cer_v0_3/cleanroom</code>	Byte-exact RFC 8785 canonicalizer with a single consumer; README states alpha, not pilot-validated, not production-certified, and other packages' canonicalizers are not yet converged on it.	31
4	Benchmark Registry Contracts	0.1.0	Contract-only	BR-1 done; BR-2 (registry, trusted resolver, revocation) not started	Digest-bound benchmark identities and typed refusals exist; every identity reports <code>trusted_resolution_performed = False</code> until BR-2 lands.	242

#	Capability	Version	Stage tag	Phase position	Where development stands	Tests
5	UVI Policy Contracts	0.1.0	Contract-only	M-2C.1 done; authority, registry and evaluator milestones deferred	Immutable policy and assessment-context shapes with structural fail-closed binding; lifecycle labels and digests remain caller-supplied inputs until a registry exists.	76
6	Policy Authority	0.2.0	Core implemented	ADR P-1 to P-11 ratified; v0.1 semantics; structured supersession pending	Issues, signs, registers, resolves and revokes policies with three external families now registered; production persistence and distributed concurrency are deferred.	279
7	Policy Workflow Compiler	0.2.0	Last phase done	Phase 1 and Phase 2 (workflow_ir.v2) complete; downstream adoption in AWC P2.1 done	Deterministic offline compiler from reviewed policy pack to Workflow IR and assurance artifacts; not pilot-validated, not production-certified; document ingestion excluded by design.	151
8	Agent Constitution Policy	0.2.0	Phase in progress	First-slice family half done (ACC-S1); first release stated to await the OD-C1=B ballot	Constitutions are issuable, signable and resolvable; README says end-to-end conformance is not yet made true, although Agentic Proposer 0.4.0 already records the OD-C1=B binding [1].	120
9	Agent Constitution Activation	0.1.0	Core implemented	ACC-IA-1 to ACC-IA-5 done with an end-to-end issue-activate-resolve-bind-conform proof	Composition root and preflight are complete; proof runs on ephemeral in-process keys and no signing key or trust root exists in the repository.	89
10	Cloud Scaling Capacity-Bounds Policy	0.1.0	Core implemented	Family adapter and rejection vocabulary done; reconciliation against Phase 5A candidates deferred	Capacity ceilings are issuable and resolvable through Policy Authority, but no composition root calls the family yet; it is not wired into any runtime path.	70

#	Capability	Version	Stage tag	Phase position	Where development stands	Tests
11	Strategy-Permission Policy	0.1.0	Core implemented	Artifact half done; concrete resolver shipped separately as capability 14	Signed strategy-permission family issued and resolved across a package boundary; a resolution proves integrity, not provenance, and authorizes no runtime action.	81
12	Agent Constitution Conformance	0.1.0	Phase in progress	Second ACC-S1-Q2 change set done; first release awaits the OD-C1=B round	Resolver and structural verifier run end to end with the family package; reference-map population remains a disclosed ungoverned gap.	103
13	Agentic Proposer	0.4.0	Core implemented	S0, S1, S2, S2-B and 0.4.0 constitution binding done; concrete evaluators and semantic auditor deferred	Contracts, identity equations and constitution binding are implemented against stubbed resolvers; classifier is pre-alpha and nothing has been exercised against a real workload.	521
14	Strategy-Permission Runtime Resolver	0.1.0	Core implemented	Ratified surface under owner rulings SURFACE=B and ROLE_LOOKUP=A	Resolves the exact signed strategy policy end to end, verified in a clean offline venv against a genuinely issued policy; role lookup exemption is test-tree-only.	90
15	Agent Workforce Composer	0.2.1	Core implemented	P1, P2 and P2.1 done; permission granting, scheduling and runtime adapters listed as next phases	Eligibility, ranking, bounded team composition and least-privilege proposals implemented offline; README records <code>pilot_validated=false</code> , <code>production_certified=false</code> .	178
16	Reasoning Method Governance	0.1.0	Research-only	Slice 1 (contracts) done	Shared research-only contracts for methods, task classes, telemetry and fit; issues no envelope and defines no approval or pilot state.	54

#	Capability	Version	Stage tag	Phase position	Where development stands	Tests
17	Reasoning Method Advisor	0.1.0	Research-only	Slice 2 done	Deterministic rule-derived advisor; every advisory is <code>RESEARCH_ONLY</code> with comparison evidence absent, and the shipped rule set is a test fixture only.	42
18	Readiness Comparison	0.2.0	Research-only	Slice 1 engine done; spec correction 30 applied	Pure comparison function with no I/O; every result is requester-asserted and research-scoped, nothing approval-bearing.	45
19	Trusted Workflow-Fit Pilot	0.1.0	Research-only	Phase 4A shipped; Phase 4C slices 1–2 landed, slice 3A/3B split in commissioning [I]	Preregistered study harness with separate-process capture and recomputed telemetry; preregistration and evaluator independence remain <code>DECLARED_UNVERIFIED</code> and approval status is constant <code>NONE</code> .	83
20	Model Selection	0.1.0	Core implemented	Model Authority rename and contract migration done; quality-floor gap from audit still open	Deterministic eligibility and selection kernel; the release is a behavior-preserving migration whose evidence remains primarily synthetic.	18
21	LLM Steering Controller	0.1.0	Core implemented	No phase ladder; provider execution outside the distribution	Advisory routing recommendations with hard-constraint filtering and reproducible evidence; README makes no claim of routing performance or production readiness.	85
22	Context Minimization	0.2.0	Core implemented	v0.1 core plus CM-TA1 token accounting done; Agent Runtime wiring in a separate package	Structural and oracle-verified minimization modes with fail-closed equivalence; carries no live-enterprise validation claim.	199

#	Capability	Version	Stage tag	Phase position	Where development stands	Tests
23	StoryGraph	2.0.0	Frozen	Frozen-but-working; legacy shim removal targeted for 3.0.0	Sequence-risk analysis with one implemented harmful graph domain; synthetic-only validation and advisory findings only.	304
24	Cloud Scaling Controller	0.4.0	Phase in progress	Phases 1–3 done in shadow mode; Phases 4–6 assigned to other packages	Canonical, predictive and cost-aware recommendations that never feed a live controller; not live-cluster validated, not production-certified.	788
25	TAP Assertion Governance Provider	0.1.0	Core implemented	Beta classifier; outcome-safety release gate in place	Working assertion-governance provider whose uncertainty-never-promoted invariant is CI-enforced; README says not production certified. Predecessor distribution was exercised by the legacy Enterprise Validation Pilot [1].	60
26	Trusted Evidence Authority	0.3.0	Core implemented	TEV-1 and TEV-2 done; BR-2, RA-5 port alignment and DD-10 production posture deferred	Verification orchestration, trust anchors, revocation, signed receipts and independent re-verification implemented; exactly one consumer imports it and production persistence and HSM posture are deferred.	530
27	Benchmark Registry Authority	0.2.3	Contract-only	BR-2A, BR-2B and BR-2C-0 done; BR-2C, BR-2D and BR-2E blocked on verifier design and DD-10	Lifecycle contracts and pure validation only; every result permanently derives <code>authority_verified is False</code> and authority-bearing types are reserved and undefined.	550
28	Risk Authority Evidence Runtime	0.1.0	Reference-grade	RA-5 complete; RA-6 to RA-8 out of this milestone	Production implementations behind Risk Authority's two ports with explicit reference versus production mode; caller-supplied PASS is inert in production mode, and HSM/KMS is excluded.	67

#	Capability	Version	Stage tag	Phase position	Where development stands	Tests
29	Cloud Scaling Producer Attestation	0.2.0	Last phase done	Phase 5B-0A complete; policy authenticity handed to 5B-0B	Producer attestations are mintable and verifiable with a gate-removal mutation sweep; <code>production_mode</code> defaults to <code>False</code> everywhere and only a reference signer ships.	324
30	Cloud Scaling Policy Authenticity	0.9.0	Phase in progress	5B-0B, 5B-1 and 5B-3 done; residual R-2 (trusted time source) left to 5B-2	Verifies the exact trusted capacity policy for tenant, scope and time; README warns that <code>as_of</code> is injected and unvalidated until envelope issuance settles the clock.	309
31	Decision Authority	1.0.0	Frozen	Public API, lifecycle, serialization and hashes frozen at 1.0.0	Bounded binding-decision kernel with no maturity caveat in its README; its legacy <code>decision_governance</code> lineage was exercised end to end by the Enterprise Validation Pilot [I].	33
32	Risk Authority	0.5.0	Phase in progress	RA-1 to RA-4 spine plus Phase 4A/4B done; Phase 5 envelope issuance and Phase 6 not implemented	Non-compensatory evaluation and v2 subject-context seam are live behind fail-closed production mode; integration stops at a non-executable <code>RiskDecision</code> and uses a reference Ed25519 implementation.	330
33	Cloud Scaling Risk Integration	0.1.0	Last phase done	Phase 4C complete; Phase 5 and 6 excluded	One-way projection into Risk Authority with recommendation-content authenticity; a fully self-consistent forgery still passes because it is not a signature.	248
34	Cloud Scaling Authorization Contracts	0.7.0	Phase in progress	Phase 5A done; 5B, 5C, 5X, 5D and 6 not implemented	Non-authoritative capacity-action candidate with measured mutation coverage; live execution remains structurally blocked until the Credential Broker phase 5X.	309

#	Capability	Version	Stage tag	Phase position	Where development stands	Tests
35	Risk Authority Runtime Composition	0.1.0	CI-verified, pilot pending	RA-4.5 composition implemented; F-D enforcement open as issue #1397	Fail-closed composition of Risk Authority, Decision Authority and ActionGate; README states production deployment validation remains pending.	62
36	ActionGate	0.1.0	Core implemented	Beta classifier; outcome-safety release gate in place	Exact-action authorization provider with CI-enforced authority invariants; README says not production certified. Predecessor distribution was exercised by the legacy Enterprise Validation Pilot [I].	57
37	Action Clearance	0.1.0	Core implemented	v0.1 core; next phases documented under the package docs/	Stateless pure-function clearance with CLEAR, HOLD, ESCALATE and BLOCK; no persistence, execution, network or domain adapters yet.	67
38	Agent Runtime	0.7.0	CI-verified, pilot pending	H22-A through H22-D done through 0.6.0; 0.7.0 current	IMPLEMENTED_AND_CI_VERIFIED lifecycle, coordination, durability and bounded concurrency; README states not live-verified, pilot-validated, distributed-safe or production-ready.	339
39	Cloud Scaling Operations	0.1.2	Core implemented	No phase ladder; DRY_RUN, SIMULATION, SHADOW and LIVE modes present	Real Kubernetes and ArgoCD actuation behind dry-run default and disabled live mode; not live-cluster validated and idempotency stores are in-memory only.	129
40	Risk Authority Status Runtime	0.1.0	Reference-grade	RA-6 operative against the ratified spec; Postgres persistence and signal transport delegated	Revocation and epoch propagation work in-process with a reference in-memory adapter; not globally consistent or zero-window revocation.	72

#	Capability	Version	Stage tag	Phase position	Where development stands	Tests
41	Risk Authority Runtime Assurance	0.1.0	Reference-grade	RA-7 done with as-built record; RA-8 handled by capability 42	Event-driven trajectory assurance that mints nothing and blocks the hot path only when opted in; reference authenticator and evaluator are refused in production mode.	96
42	Risk Authority Execution Assurance	0.1.0	Reference-grade	RA-8 done including the M-1 non-compensatory closure; last milestone of the RA-5 to RA-8 ladder	Post-effect reconciliation emitting evidence and a neutral reassessment signal; content-hash integrity only, with no signed external receipts or third-party gateway.	143
43	Context-Minimization Token-Accounting Runtime	0.1.0	Core implemented	CM-TA1 integration; no numbered successor	One-way bridge from Agent Runtime telemetry to accounting records and budget settlement; no real provider adapter and only an in-memory reference sink.	65
44	Agent Value Readiness	0.4.1	Experimental kernel	M-3R.1 to M-3R.3 done plus Trusted Readiness Orchestration; ROI, forecasting and deployment authorization deferred	Deterministic three-dimension readiness determination marked experimental, internal, advisory and non-financial; no allow-all verifier ships by design.	542
45	Governed Value	0.2.0	Experimental kernel	GV-0 and GV-1 done; GV-2 evidence and GV-4 authority layers do not exist	Reported-value calculation kernel over caller-reported inputs; every figure carries REPORTED evidence status and UNVERIFIED authority status.	42

B.5 Distribution by stage

Stage tag	Count	Capabilities
Core implemented	19	1, 2, 3, 6, 9, 10, 11, 13, 14, 15, 20, 21, 22, 25, 26, 36, 37, 39, 43
Phase in progress	6	8, 12, 24, 30, 32, 34
Reference-grade	4	28, 40, 41, 42
Research-only	4	16, 17, 18, 19
Contract-only	3	4, 5, 27
Last phase done	3	7, 29, 33
CI-verified, pilot pending	2	35, 38
Experimental kernel	2	44, 45
Frozen	2	23, 31
Pilot-ready	0	none
Pilot-validated	0	none
Production-certified	0	none

B.6 Reading the table

1. **The accountability chain is implemented but not piloted.** Every Core-required capability from Appendix A is at least Core implemented, and the four Risk Authority runtime packages are reference-grade with production adapters delegated rather than absent. What is missing is a bounded pilot of the current package lineage; the only pilot evidence is the legacy Enterprise Validation Pilot over predecessor distributions.
2. **Cloud scaling is the deepest phase ladder and the clearest gap.** Phases 1–5B are landed across six packages, but envelope issuance (5B-2), production ActionGate admission (5C), the Credential Broker (5X), bounded execution (5D) and effect verification (6) are all named and unbuilt. Live infrastructure mutation is structurally blocked until 5X.
3. **Three capabilities are contracts without engines.** Benchmark Registry Contracts, UVI Policy Contracts and Benchmark Registry Authority ship typed shapes whose authority fields are permanently false by construction. They should not be presented as operational registries or authorities.
4. **The reasoning-method thread is research-only end to end.** Capabilities 16–19 are commissioned under an owner-ratified research ballot and label every output non-approval-bearing; Phase 4C of the workflow-fit pilot is actively landing but does not change that label.
5. **Version numbers and README phase text drift in places.** Cloud Scaling Authorization Contracts is at 0.7.0 while its README narrates 0.1.0 to 0.2.0; Cloud Scaling Controller's header says 0.3.0 while its Phase 3 section says 0.4.0; Agent Runtime's Status section stops at 0.5.0 while its Maturity block reaches 0.6.0 and the distribution is 0.7.0. The Version column above reports the distribution's `__version__` at the inspected commit.

Appendix C — Cross-Check Log and Competitor Supplement

C.1 Method

Sixteen factual claims in Sections 4–13 and Appendix A were checked against package source and READMEs under `packages/` at the inspected commit, following the repository rule that code and package documentation outrank explanatory prose. Verdicts use the working-agreement labels: [V] verified, [I] inferred, [G] gap. Where a claim was contradicted or partial, the body text was corrected in place and the correction is recorded here so the change is auditable.

C.2 Claim-by-claim results

#	Claim in document	Verdict	Evidence	Action taken
1	Package count of 47 and two excluded business-solution packages	[V]	47 <code>pyproject.toml</code> under <code>packages/</code> ; <code>packages/products/</code> holds <code>ai-hiring</code> and <code>procurement</code>	None
2	ActionGate returns an <code>ALLOW</code> outcome meaning the exact action matches authority (A.2.6)	Contradicted on naming	Native <code>ActionGateOutcome</code> is <code>ALLOW</code> / <code>DENY</code> / <code>ALLOW_WITH_CONSTRAINTS</code> / <code>UNKNOWN</code> in <code>providers/actiongate/.../core.py</code> ; the public framework outcome is <code>AUTHORIZED</code> / <code>AUTHORIZED_WITH_CONSTRAINTS</code> / <code>DENIED</code> / <code>INDETERMINATE</code> in <code>mapping/result.py</code>	A.2.6 and §36 corrected to the framework outcome names
3	TAP classifies assertions as supported, unsupported, constrained or indeterminate (§25)	Partial	Native <code>TapOutcome</code> adds <code>UNKNOWN</code> , mapped fail-safe to <code>INDETERMINATE</code> (<code>providers/tap/.../core/__init__.py</code> , <code>mapping/result.py</code>)	§25 now names the fifth native state
4	Action Clearance statuses and precedence (§37, A.2.7)	[V]	<code>ClearanceStatus</code> and <code>STATUS_PRECEDENCE</code> in <code>action-clearance/.../models/enums.py</code>	None
5	Risk Authority converts a decision into signed, scoped, time-bound, revocable machine authority (§32)	Partial, materially	<code>EnvelopeIssuer</code> , Ed25519 signing, 30-minute default TTL and revocation exist; in <code>production_mode</code> issuance and authorization raise <code>ProductionContainmentError</code> as Phase 5 work (<code>risk_authority/README.md</code> , <code>api/dependencies.py</code>)	§32 now states the reference-path versus production-mode split
6	Agent Constitution Policy states permitted dispositions, review actions and tool scopes (§8)	[V]	Three bound fields on <code>AgentConstitutionPolicy</code> in <code>agent-constitution-policy/.../policy.py</code>	None

#	Claim in document	Verdict	Evidence	Action taken
7	StoryGraph is deterministic, policy-defined and advisory (§23)	[V]	README authority boundary; determinism stated in <code>storygraph/.../model.py</code>	None
8	Governed Value calculates value per authorized action using authority classification (§45)	Contradicted	Scorer pins <code>EvidenceStatus.REPORTED</code> and <code>AuthorityStatus.UNVERIFIED</code> (<code>governed-value/.../services/scorer.py</code>); no authorization or envelope reference exists in the package	§45 corrected; A.2.10 and A.5 row 45 remain as stated design intent for the illustrative scenario
9	Cloud Scaling Operations performs Kubernetes or ArgoCD operations with dry-run default (§39)	[V] with omission	Executors exist; <code>ExecutionMode.DRY_RUN</code> is default among <code>DRY_RUN</code> , <code>SIMULATION</code> , <code>SHADOW</code> , <code>LIVE</code> (<code>contracts.py</code>)	§39 now names the four modes
10	Model Selection selects an eligible model under policy (§20)	Stale framing	Package brands itself Model Authority; <code>ModelAuthorizationDecision</code> with <code>ALLOW / DENY / HOLD / ESCALATE</code> is canonical (<code>model-selection/.../authority.py</code>)	§20 now records the Model Authority contract
11	Agent Runtime fails closed when governance is not configured (§38)	[V]	<code>UnconfiguredGovernanceHook</code> default in <code>agent-runtime/.../config.py</code> ; fail-closed branches in <code>runtime/engine.py</code>	None
12	Context Minimization preserves a caller-defined equivalence condition with token accounting (§22)	[V]	<code>EquivalenceStatus</code> , <code>OracleRequiredError</code> , <code>token_accounting.py</code>	None
13	Policy Workflow Compiler emits Workflow IR, assurance specs, capability requirements, audit schemas and content-addressed artifacts (§7)	[V]	<code>CompilationResult</code> fields and <code>CapabilityRequirement</code> in <code>policy-workflow-compiler/.../compiler.py</code> , <code>semantics/contracts.py</code>	None
14	Agentic Proposer may propose, request evidence, abstain or escalate (§13)	[V]	<code>CandidateDisposition</code> in <code>agentic-proposer/.../vocabulary.py</code>	None
15	Agent Workforce Composer explains eliminated candidates and proposes least-privilege permissions and fallbacks (§15)	[V]	<code>EliminationReason</code> , <code>permissions.py</code> , <code>fallback.py</code>	None
16	Decision Authority records delegated authority with scope, conditions and validity (§31)	[V]	<code>AuthorityType.DELEGATED_POLICY</code> , <code>decision_scope</code> , <code>expires_at</code> in <code>decision-authority/.../services/</code>	None

Two further observations from the check, not tied to a single claim:

- **Version drift between READMEs and distributions** was already recorded in B.6 and is not repeated here.
- **The Enterprise Validation Pilot** cited in B.1 ran over predecessor distributions ([decision-governance](#), [dgm-tap-provider](#), [dgm-actiongate-provider](#)). The current packages keep those names as compatibility distributions, so the lineage is documented, but the pilot result is [\[I\]](#) for the current code, not [\[V\]](#).

C.3 Competitor supplement

Thirteen entries in A.7 previously carried an empty competitor field: 1, 3, 9, 11, 13, 14, 15, 17, 22, 26, 27, 30 and 34. Each now cites the nearest product or open standard found, with the boundary of the overlap stated. Twenty-six further entries received an "Additional analogues" line where the original review cited only one vendor family. A.9 lists the new vendor groups.

Sourcing rules for the supplement:

- Every cited page was located through web search on the day of the check and is an official vendor, documentation or standards page. Direct page fetches were blocked by the session's network policy for most vendor domains, so page content was confirmed from search excerpts rather than a full read. Descriptions are therefore kept to what those excerpts support.
- Open-source projects and open standards (for example DSPy, Presidio, Karpenter, MCP, A2A, SPIFFE) are labelled as such in the text and are included because they are the closest functional analogue, not because they are commercial competitors.
- A cited analogue overlaps one function of a capability. None was found to reproduce the platform's separation of proposal, verification, decision, machine authority, exact-action authorization, clearance, execution, assurance and value measurement, and the caveat in A.9 about blank fields applies equally to the supplement.